

Tracking the Credibility Revolution across Fields*

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Abstract

How far has the credibility revolution spread beyond applied microeconomics? I revisit Currie, Kleven, and Zwiers (2020b) using about 39,400 papers: 27,100 NBER working papers from 2000–2025 and 12,300 articles from eleven top economics and finance journals from 2011–2024. I track method diffusion through keyword mentions. Three results emerge. First, finance, macro, and other fields lag applied micro. In 2024, 65 percent of applied micro papers mention experimental or quasi-experimental methods, versus 51 percent in finance and 47 percent in macro and other fields. Second, the spread outside applied micro is mostly difference-in-differences (DiD). Including DiD raises the share of finance papers mentioning any experimental or quasi-experimental method by about 53 percent, compared with 29 percent in applied micro. Other methods—instrumental variables, regression discontinuity, and experiments—show much less growth. Third, I document a striking gap between the methods studied in the *Journal of Econometrics*—where nonparametric estimation and asymptotic theory dominate—and those used by applied researchers, where DiD and identification strategies dominate. Published journal articles confirm these patterns are not artifacts of the NBER working paper sample.

JEL Codes: C18, C81, B41

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How far has the credibility revolution spread? Angrist and Pischke (2010) documented a sea change in how economists approach empirical work—a shift toward transparent research designs, explicit identification strategies, and credible causal inference. Currie, Kleven, and Zwiers (2020b) showed that this shift was accelerating through the late 2010s, at least in applied microeconomics. But that analysis left open a basic question: are finance, macroeconomics, and other fields keeping pace, or has the revolution been narrower than it appears?¹

I take up this question by extending Currie, Kleven, and Zwiers (2020b)’s approach to a much larger corpus. Using keyword matching on the full text of about 39,400 papers—27,100 NBER working papers (2000–2025) and 12,300 articles from eleven top journals (2011–2024)—I track mentions of empirical methods across fields and over time. I collect NBER full text back to 1982, but use the pre-2000 papers only to benchmark coverage against Currie, Kleven, and Zwiers (2020b); the main analysis begins in 2000. The expanded sample adds finance and macro/other fields, which were omitted from the original analysis, and supplements working papers with published journal articles. Because the analysis measures keyword *mentions* rather than verified *use*, the trends should be interpreted as tracking the diffusion of methodological language—a proxy for, but not identical to, actual method adoption.

The analysis shows that mentions of credibility revolution methods have spread unevenly across fields. I organize the findings around three main results.

First, finance and macro/other differ substantially from applied micro on most measures. As of 2024, 65 percent of applied micro papers mention experimental or quasi-experimental methods, compared to 51 percent in finance and 47 percent in macro/other. In identification language, the current levels in finance and macro/other are comparable to where applied micro was around 2009–2010. The gap has narrowed but remains visible.

Second, the credibility revolution outside applied micro has been—to a first approximation—a difference-in-differences revolution. Including DiD in the methods measure raises the finance share by roughly 53 percent versus 29 percent for applied micro. Other quasi-experimental tools—instrumental variables, regression discontinuity, RCTs—have seen far less growth in finance and macro. This reliance on a single method is striking given the recent econometrics literature highlighting sensitivities in DiD designs (Roth 2022; De Chaisemartin and d’Haultfoeuille 2020; Callaway, Goodman-Bacon, and Sant’Anna 2024).

Third, I document a pronounced gap between the methods studied in the *Journal of Econometrics*—where nonparametric estimation, bootstrap methods, and asymptotic theory dominate—and those used by applied researchers, where DiD and identification strategies are the dominant tools. The tools powering the credibility revolution and the theoretical literature developing new estimators occupy largely separate methodological spaces.

Two features of the analysis strengthen confidence in these patterns. Published articles from top journals show trends that closely mirror the NBER data, with slightly higher rates of credibility revolution methods—consistent with a publication selection effect favoring papers using causal

¹Throughout this paper, I use “macro/other” to refer to the NBER field grouping that includes macroeconomics alongside several other programs; see Tables 1 and 2.

inference methods for identification. And a validation exercise using LLM-based classification confirms that keyword matching achieves 72–92 percent agreement rates for most method categories with more sophisticated approaches at near-zero computational cost, though agreement is lower for broader categories like identification strategy and structural models.

This paper is complementary to recent work by Brodeur, Cook, and Heyes (2020), who hand-code a sample of 684 papers (2015–2018) to classify methods, and Garg and Fetzer (2025), who use LLMs to classify causal inference methods in a sample of 44,852 papers. Many of the aggregate trends I document are consistent with Garg and Fetzer (2025)’s findings, although their work is much more carefully documenting the actual causal claims themselves. I contrast our findings in more detail in Appendix A.

The paper proceeds as follows. Section 1 describes the data and methods. Section 2 presents the NBER working paper results, documenting trends across fields, programs, and methods. Section 3 extends the analysis to published articles from top journals. Section 4 examines the gap between econometric theory and applied practice. Section 5 concludes.

1 Data and Methods

I measure mentions of empirical methods over time following the approach in Currie, Kleven, and Zwiers (2020b): searching the full text of papers for keywords and regular expressions that capture the language of the credibility revolution (e.g. “threats to identification” or “identification strategy”).²

NBER Working Papers. I collect the full text of approximately 31,500 NBER working papers from the NBER website, covering working papers numbered 1000 through the most recent available (1982–2025). The main analysis uses the 27,100 papers from the year 2000 onward; the pre-2000 papers are used only to compare coverage and historical trends with Currie, Kleven, and Zwiers (2020b). Unlike Currie, Kleven, and Zwiers (2020b), who focus exclusively on “applied micro” papers, I include all papers in the NBER working paper series. Each paper is associated with one or more of nineteen NBER research programs, which I use for field classification.

Top Journal Articles. I supplement the NBER data with articles from eleven leading economics and finance journals, covering 2011–2024: three general-interest economics journals (AER, QJE, JPE), the four American Economic Journals (Applied, Policy, Macro, Micro), three top finance journals (Journal of Finance, Review of Financial Studies, Journal of Financial Economics), and the Journal of Econometrics. I extract full text from published PDFs using PyMuPDF. For the AER, I filter out Papers and Proceedings (P&P) articles using DOI patterns: issue-based identification for 2011–2014 (when P&P appeared in a designated issue of the AER) and DOI prefix matching for 2015–2017 (when P&P received distinct DOI prefixes); from 2018 onward, P&P papers moved to a separate journal. Appendix K documents text extraction rates by journal and year. In total, the journal sample comprises approximately 12,300 articles.

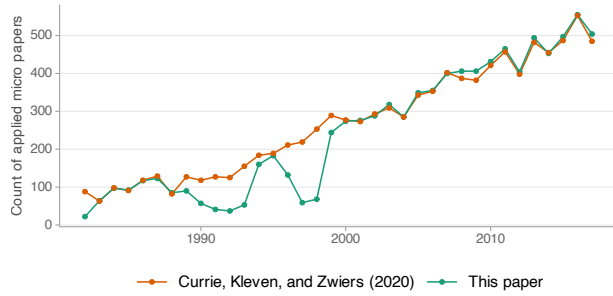
²See the Appendix for the full set of keywords. I follow the same method as Currie, Kleven, and Zwiers (2020b).

Text Processing. For each paper, I extract the full text, strip out the references section—identified by looking for section headers followed by high concentrations of “Journal” mentions—and apply the keyword search. I use the same keywords and regular expressions as Currie, Kleven, and Zwiers (2020b), with appropriate case sensitivity for each category. The full list is in the Appendix. Full-text search of the papers used in this manuscript are available at <https://paulgp.com/econlit-pipeline/> to allow for alternative keyword searches.

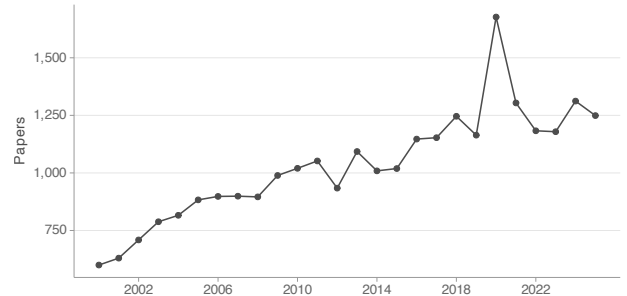
Validation. I validate keyword matching against two external benchmarks in Appendix A. First, I compare keyword flags to the hand-coded method labels in Brodeur, Cook, and Heyes (2024), matching 484 papers across nine journals (2002–2020) by title. Treating the hand-coded labels as ground truth, keywords achieve high recall—97% for DiD and IV, 95% for RD, 89% for RCT—meaning they rarely miss a paper that uses a given method. Precision is lower (57–69%), reflecting that keywords also flag papers that mention a method without using it as a primary research design. Second, I classify a stratified sample of 750 papers using two independent LLMs (Claude Haiku 4.5 and Qwen 3.6-35B). The two LLMs produce nearly identical positive rates for most method categories (within one to two percentage points), diverging mainly for event studies and administrative data, and keyword–LLM accuracy (the share of papers where the keyword flag agrees with the LLM) runs 72–92% for most categories—though agreement is lower for identification strategy (58%) and structural models (67%). LLM-keyword *precision*—the share of keyword-flagged papers the LLM confirms as actually using the method—is much lower (below 50%) across nearly all method categories, reflecting that keyword matching flags any mention while the LLM treats only methods used in the research design as positives.

Field Classification. For journal articles, I classify papers into fields using a two-step procedure. First, field-specific journals are directly classified: AEJ Applied and AEJ Policy map to “Applied Micro,” AEJ Macro to “Macro,” AEJ Micro to “Micro Theory,” the three finance journals (JF, JFE, RFS) to “Finance,” and the Journal of Econometrics to “Econometrics.” Second, for the general-interest journals (AER, QJE, JPE), I use JEL codes when available. Each paper’s JEL code first letters determine its field: D, J, L, H, I, Q, R, or K codes map to “Applied Micro”; G codes to “Finance”; E or F codes to “Macro”; and C codes to “Econometrics.” When a paper has JEL codes spanning multiple fields, I assign it to the first matching field in the priority order listed above—applied micro takes precedence, then finance, then macro, then econometrics. This reflects the fact that many papers cross boundaries (e.g., a paper with both D and G codes is likely applied work using financial data). Papers without JEL codes—primarily from QJE and JPE, which do not report them—default to “General Econ.” In the AER, 87% of papers are classified as Applied Micro via JEL codes, 5.7% default to General Econ, and the remainder split across Finance, Macro, and Econometrics.

As Currie, Kleven, and Zwiers (2020b) note in their replication package (Currie, Kleven, and Zwiers 2020a), PDF-to-text conversion introduces errors. To see how this affects my sample, I compare paper counts over time in the “applied micro” setting to Currie, Kleven, and Zwiers (2020b)



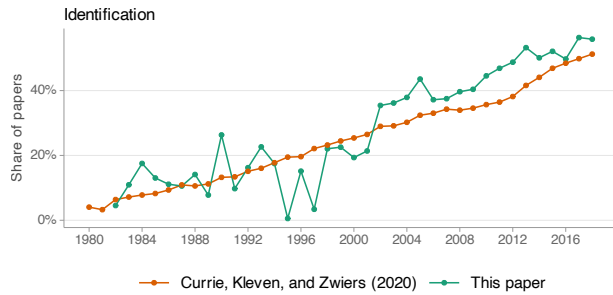
(i) Comparison of sample size to Currie, Kleven, and Zwieters (2020b) in “applied micro”



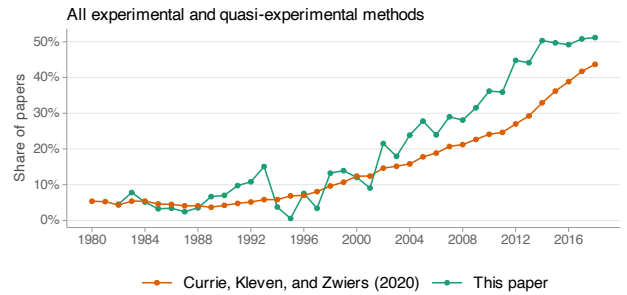
(ii) Total papers in final sample over time

Figure 1: NBER Working Paper Counts over Time. Data for Currie, Kleven, and Zwieters (2020b) is measured in Appendix Figure B.I. in their paper. My sample ends in early 2025.

in Figure 1(i). My sample has more gaps in the 1990s—reflecting data processing errors for PDFs in that period—but coverage is close in the early 1980s and from 1999 onwards. Figure 2 provides a more direct check: I compare two headline estimates from Currie, Kleven, and Zwieters (2020b) to mine—the share of papers referencing identification and the share referencing any experimental or quasi-experimental method (RCTs, lab experiments, difference-in-differences, regression discontinuity, instrumental variables, event studies, or bunching). My estimates track well except in the late 1990s. I therefore use the pre-2000 observations only for this comparison and focus the main analysis on 2000 onward, leaving 27,078 NBER working papers. Figure 1(ii) plots the sample over time.



(i) Comparison of identification measure to Currie, Kleven, and Zwieters (2020b) in “applied micro”



(ii) Comparison of all experimental and quasi-experimental methods measure to Currie, Kleven, and Zwieters (2020b) in “applied micro”

Figure 2: Validation of measurement with Currie, Kleven, and Zwieters (2020b). Data for Currie, Kleven, and Zwieters (2020b) is taken from Figure 2 Panel A and B. I plot the raw (annual) measure, while the Currie, Kleven, and Zwieters (2020b) data is a rolling five-year mean; the smoothing explains the slight visual discrepancy between the two series.

Each NBER working paper can be submitted to one or more of nineteen programs, and 61 percent list more than one.³ Table 1 reports the breakdown. The three largest programs are Economic

³39 percent have one program, 33 percent have two, 18 percent have three, 7 percent have four, and 2 percent have five.

Fluctuations and Growth (macroeconomics), Public Economics (applied micro, as classified by Currie, Kleven, and Zwiars (2020b)), and Labor Studies (also applied micro). Development Economics is smaller in part because it began only in 2012.

NBER Program	Number of Papers
Applied Micro	
Labor Studies	5,970
Public Economics	5,896
Economics of Health	3,641
International Trade and Investment	2,466
Children and Families	2,193
Industrial Organization	2,160
Economics of Education	2,105
Development Economics	1,955
Political Economy	1,869
Environment and Energy Economics	1,724
Economics of Aging	1,698
Finance	
Asset Pricing	2,985
Corporate Finance	2,785
Macro/Others	
Economic Fluctuations and Growth	5,645
International Finance and Macroeconomics	3,107
Monetary Economics	2,924
Productivity, Innovation, and Entrepreneurship	2,785
Development of the American Economy	1,675
Law and Economics	1,385

Table 1: NBER Working Paper Series counts by program

To compare across programs, I extend Currie, Kleven, and Zwiars (2020b)’s classification. I define “finance” as Asset Pricing and Corporate Finance, and “macro/other” as the remaining programs: Development of the American Economy (the economic history group), Economic Fluctuations and Growth, International Finance and Macroeconomics, Law and Economics, Monetary Economics, and Productivity, Innovation, and Entrepreneurship. Table 2 reports the resulting field counts. One difference from Currie, Kleven, and Zwiars (2020b) is worth noting: they define applied micro as papers that *solely* list applied micro programs, making it a paper-specific label. I instead use non-exclusive labels—if a paper is in both finance and applied micro, it counts in both categories. Among post-2000 papers with at least one field classification, 44 percent are exclusively applied micro, seven percent exclusively finance, and 19 percent exclusively macro/other. The cross-field groups are applied micro plus macro/other (19%), finance plus macro/other (6%), applied micro plus finance (2%), and all three fields (3%). Appendix D reports all main results under an exclusive classification (papers assigned to a single field); the cross-field gaps are qualitatively similar and, if anything, slightly wider.

Throughout the analysis, field and program labels are non-exclusive: a paper contributes to every program and every field to which it is submitted. The collected corpus includes approximately 31,500 NBER papers and 12,300 journal articles. The main NBER analysis uses 27,078 papers from 2000 onward, of which 26,849 have at least one of the field labels in Table 2. These papers generate 35,793

Field Combination	Number of Papers
Applied Micro only	11,828
Finance only	1,758
Macro/Others only	5,111
Applied Micro + Finance	577
Applied Micro + Macro/Others	5,091
Finance + Macro/Others	1,692
All three	792
Total field-classified papers	26,849

Table 2: Non-exclusive NBER field memberships. Counts are post-2000 papers with at least one program in each field; rows do not sum to the number of papers because cross-listed papers contribute to each field.

paper-field observations because field membership is non-exclusive. The pre-2000 NBER papers are used only when comparing coverage and historical trends with Currie, Kleven, and Zwiers (2020b).

Table 3 provides a snapshot of the headline numbers. For each field, I report the share of papers mentioning identification, experimental or quasi-experimental methods (with and without DiD), and DiD specifically, both for 2016–2024 and for the 2000–2015 average. These numbers summarize the main patterns documented in the paper.

Field	2016–2024					2000–2015				
	<i>N</i>	Ident.	Exp./QE	DiD	Excl. DiD	<i>N</i>	Ident.	Exp./QE	DiD	Excl. DiD
Applied Micro	8,265	50.8%	61.4%	29.8%	47.7%	9,067	39.8%	45.2%	14.7%	38.4%
Finance	2,089	38.8%	45.1%	26.6%	29.4%	2,517	25.9%	27.4%	13.2%	18.1%
Macro/Others	5,355	40.2%	43.8%	21.7%	31.7%	6,714	28.8%	29.1%	9.6%	23.7%

Table 3: Summary of credibility revolution measures by field. Shares are computed from NBER working papers using non-exclusive field membership, so cross-listed papers contribute to each field. “Exp./Quasi-exp.” includes DiD, event studies, IV, RD, RCTs, lab experiments, and bunching. “Exp./QE excl. DiD” excludes difference-in-differences and event studies.

2 Results from NBER Working Papers

2.1 Overall trends

Figure 3 presents the updated version of Currie, Kleven, and Zwiers (2020b)’s Figure 2 for the post-2000 NBER sample, through early 2025.⁴ Each panel shows field-specific trends as colored lines, with the overall aggregate as a dashed black line.

Nearly all trends continue in the direction Currie, Kleven, and Zwiers (2020b) documented. The share of papers explicitly mentioning identification has risen overall, with growth slowing markedly since 2016 (panel a).⁵ The share mentioning any experimental or quasi-experimental method, by

⁴Currie, Kleven, and Zwiers (2020b) use a five-year moving average; I present three-year moving averages throughout.

⁵The keywords are listed in the Appendix, as well as in the Appendix of Currie, Kleven, and Zwiers (2020b). For example, matches for identification flag phrases like “identification assumption” and “causal identification.”

contrast, has continued to rise even after 2016 (panel b). This means identification language has saturated while mentions of specific methods keep growing. Administrative data (panel c) has also continued its upward trend.

But the aggregate trends mask substantial heterogeneity. Figure 3 previews the paper’s central finding: mentions of credibility revolution methods have spread unevenly, with persistent gaps between applied micro on the one hand and finance and macro/other on the other. The next two subsections document these gaps in detail and decompose them by method and program.

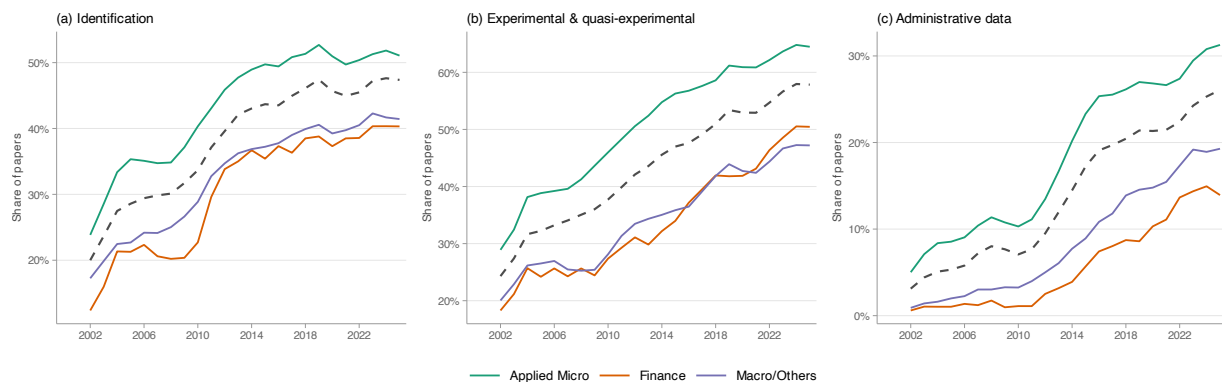


Figure 3: Credibility revolution trends in NBER working papers (three-year moving averages). Colored lines show field-specific trends; dashed black line shows the overall aggregate. See Table 2 for field definitions and the Appendix for keyword definitions.

Graphical revolution trends (figures-to-tables ratio) are reported in Appendix B.

2.2 Comparison across fields

Figure 3 splits each variable by the three field groupings defined in Table 2. The gaps are large and persistent. For identification, experimental and quasi-experimental methods, and administrative data, applied micro is well above both finance and macro/other. Applied micro’s identification share has grown more slowly since 2017, reaching 52 percent by 2024, and remains 10–11 percentage points above finance and macro/other. For experimental and quasi-experimental methods (panel b), applied micro reaches 65 percent by 2024, while finance stands at 51 percent and macro/other at 47 percent.⁶ All time-series figures use three-year moving averages, which smooth year-to-year fluctuations; averages for 2016–2024 are reported in Table 3. These cross-field gaps also rely on non-exclusive field labels (a paper listing applied micro and finance programs counts in both); Appendix D replicates the analysis using exclusive labels and finds the gaps qualitatively similar and, if anything, slightly wider.

To put these gaps in context, it helps to ask where finance and macro/other stand *today* relative to applied micro in the *past*. In identification, the current levels in finance and macro/other are

⁶The “macro/other” grouping pools the Economic Fluctuations and Growth, International Finance and Macroeconomics, and Monetary Economics programs (core macro) with Law and Economics, Productivity/Innovation/Entrepreneurship, and Development of the American Economy. These are not methodologically homogeneous; Figures 6 and 7 report program-level results separately.

comparable to applied micro circa 2009–2010. In experimental and quasi-experimental methods, finance is comparable to applied micro circa 2012 and macro/other to applied micro circa 2010–2011. In administrative data, both are comparable to applied micro circa 2013. Whether this reflects a lag that will eventually close or different long-run equilibria—driven by differences in available data and the nature of the research questions—is an important open question.

Figure 4 presents method-specific trends by field. I start with difference-in-differences (panel a), which includes event studies. All three fields show steep growth, with applied micro leading. Finance is close behind—partly because the term “event study” captures financial event studies (abnormal return studies) that differ methodologically from DiD-style event studies. Appendix C decomposes this measure into strict DiD language and event study mentions, and identifies papers combining “event study” with “abnormal return” as a proxy for financial event studies. While financial event studies account for a meaningful share of finance’s event study mentions, they do not explain the full convergence: finance’s strict DiD share has also grown substantially.

Panel (b) tells a different story: synthetic controls. In Appendix Figure A.V of Currie, Kleven, and Zwiers (2020b), synthetic control was growing rapidly as of 2018 among applied micro papers. That growth continued through 2020 but has since leveled off. Even when including “synthetic DiD” methods (Arkhangelsky et al. 2021) and related variants, the combined measure shows that synthetic control mentions have plateaued—a notable contrast to DiD’s continued rise. Appendix H decomposes classic synthetic control from synthetic DiD mentions.

I also examine Bartik and shift-share instruments (Goldsmith-Pinkham, Sorkin, and Swift 2020; Borusyak, Hull, and Jaravel 2022; Adão, Kolesár, and Morales 2019) (panel c). Since 2013, this method has grown rapidly across all fields, though with some decline in macro/other and finance after 2021. As of 2024, 4–7 percent of papers mention Bartik or shift-share. To see what this means in context, panel (d) plots the share mentioning instrumental variables at all. That share has stayed roughly constant over time—at about 27 percent for applied micro, 19 percent for finance, and 21 percent for macro/other. As a back-of-the-envelope calculation, dividing the Bartik/shift-share share by the IV share gives a marginal ratio of about 19–28 percent. Reading this as a conditional probability requires assuming that papers mentioning Bartik or shift-share also mention IV, which is plausible since these are themselves IV strategies but is not verified directly here.

Turning from instruments to experiments and regression discontinuity: in panel (e), applied micro leads in RCT mentions, with 23 percent of papers by 2024. Finance and macro have also grown, but more slowly. The divergence is striking because all three fields started from low bases as of 2003—this means the gap is largely a post-2003 phenomenon. In panel (f), applied micro leads finance and macro/other by about 5–6 percentage points in regression discontinuity mentions as of 2024, but the share has flattened across all fields over the past eight years. RD appears to have reached a natural ceiling.

What accounts for the gap between applied micro and the other fields? Some papers may be pure theory or observational work. Another possibility—already measured by Currie, Kleven, and Zwiers (2020b)—is structural estimation (keywords include “structural model,” “structural general equilibrium model,” and “GMM”). In Figure 5(i), macro/other and finance have a 2–3 percentage

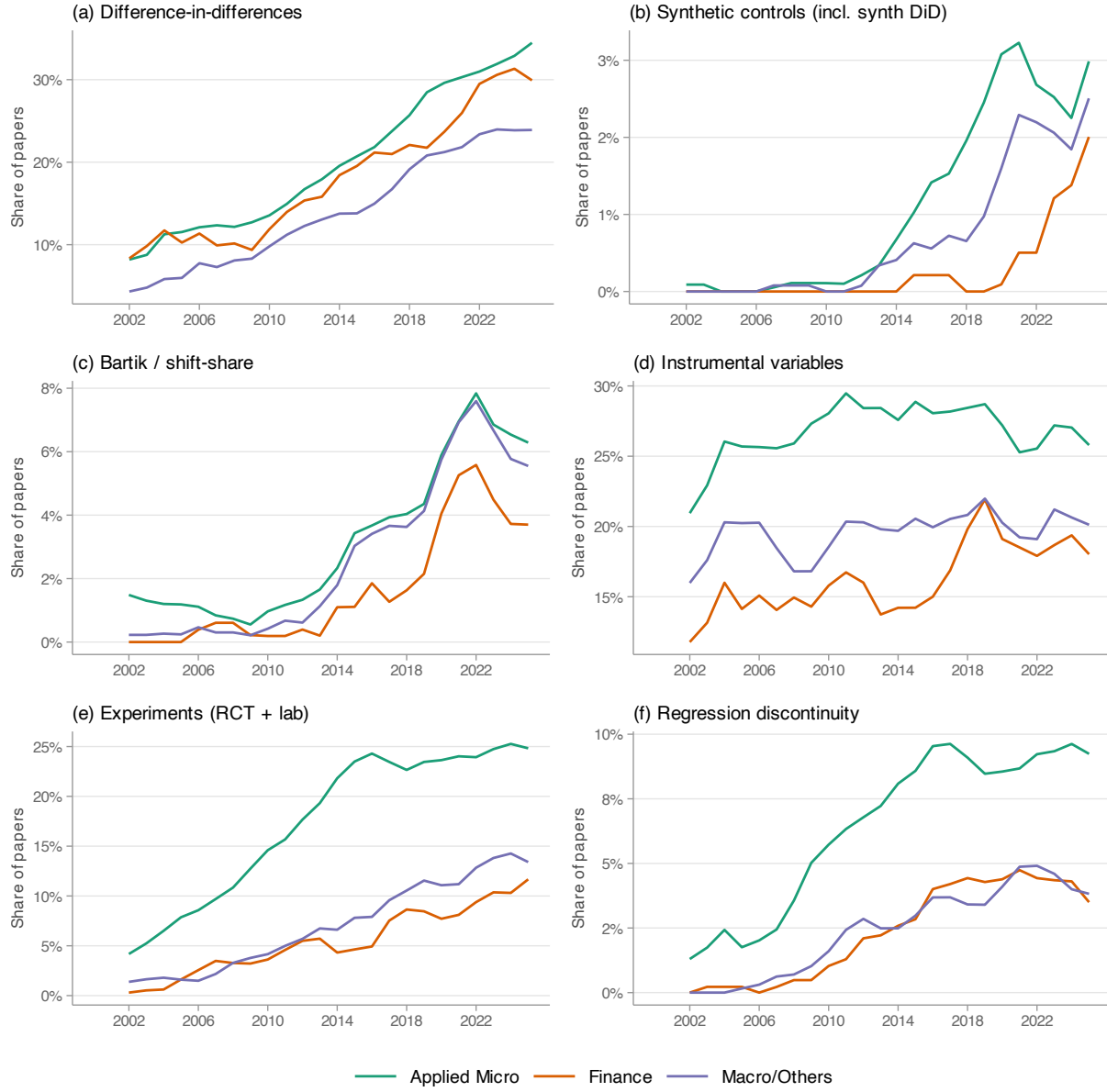
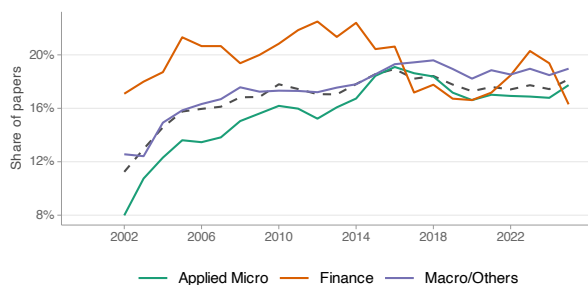


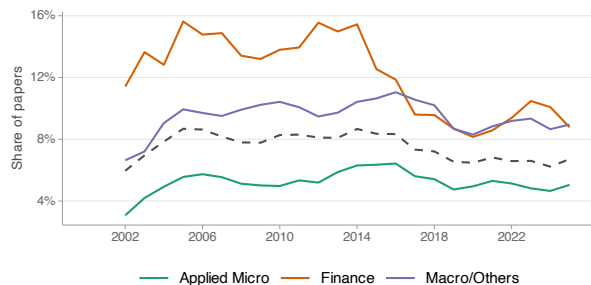
Figure 4: Method-specific trends by field (three-year moving averages). Note: y-axis ranges differ across panels to accommodate different prevalence levels. See Table 2 for field definitions and the Appendix for keyword definitions.

point higher share of structural estimation mentions, consistent with the greater prevalence of structural models in these fields (though applied micro includes Industrial Organization, which also relies heavily on structural methods). Because the broad measure includes GMM and MLE—which appear in many non-structural contexts—Appendix F also reports a narrow measure excluding these terms. The cross-field pattern is qualitatively similar under both definitions.

More revealing is Figure 5(ii), which isolates papers that mention structural estimation *without* also mentioning experimental or quasi-experimental methods. Here the gap widens: finance and macro/other papers are roughly twice as likely to fall in this category as applied micro papers as of 2024. This pattern is consistent with applied micro papers more often pairing structural language with quasi-experimental language, though it does not by itself establish that the finance and macro papers in this category lack identification strategies—only that they do not also mention the experimental/quasi-experimental keywords. Structural estimation thus accounts for part of the cross-field gap in the keyword measures. More broadly, part of the gap reflects denominator composition: finance and macro/other have a larger share of purely theoretical papers. Conditioning on papers that mention at least one empirical method or data source narrows the gap—the experimental/quasi-experimental share rises to 78 percent for applied micro, 69 percent for finance, and 66 percent for macro/other in 2024—but the gap persists (Appendix G). One caveat about this exercise: the “empirical” denominator is constructed from the same keyword universe as the headline measures, so it cannot fully separate denominator composition from differences in the vocabulary that empirical papers use.



(i) Structural Models



(ii) Structural Models without mention of experimental or quasi-experimental methods

Figure 5: Panel (a) reports the share of papers that mention structural model estimation. Panel (b) reports the share of papers that mention structural model estimation and do not mention any form of experimental or quasi-experimental methods. Colored lines show field-specific trends; dashed black line shows the overall aggregate. See Table 2 for the breakdown of fields, and the Appendix for definitions on keywords.

2.3 Breakdown across programs

The field-level averages mask important within-field variation. Within finance, asset pricing and corporate finance produce very different types of empirical work. The same is true within applied micro and macro/other. To see what is driving the field-level trends, I disaggregate to individual NBER programs.

Figure 6 plots the share of papers mentioning identification and experimental/quasi-experimental methods across all nineteen programs using slope charts. Each line segment connects a program’s 2000–2015 share (left) to its 2016–2024 share (right), colored by field. The slope and level of each segment simultaneously reveal where programs stand and how much they have changed.

Despite within-field heterogeneity, the cross-field pattern is strikingly consistent. In panel (a), applied micro programs have higher identification shares than nearly all finance and macro/other programs, with the exception of Law and Economics. Within finance, there is a large gap between Asset Pricing and Corporate Finance—this means the rise of identification in finance is driven primarily by Corporate Finance. In panel (b), the pattern is similar: applied micro programs generally lead, with Law and Economics, Corporate Finance, and Productivity, Innovation, and Entrepreneurship among the non-applied-micro programs reaching comparable levels. Identification mentions grew at broadly similar rates across programs, while growth in experimental and quasi-experimental methods varies sharply: International Finance and Macroeconomics and Economic Fluctuations and Growth have seen markedly less growth than Corporate Finance and Law and Economics.

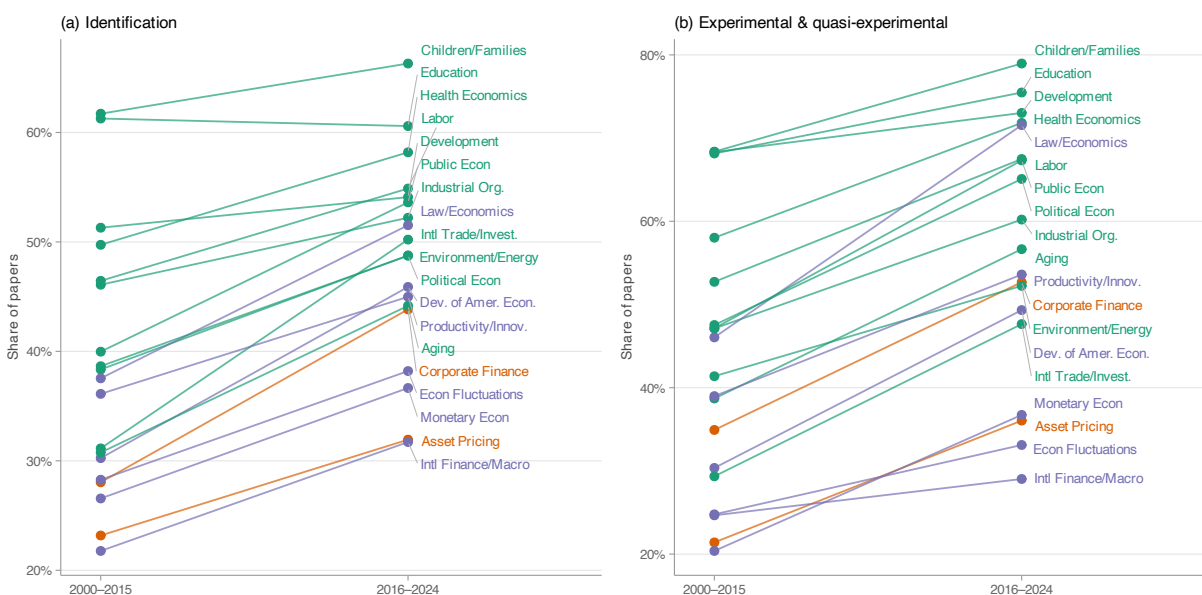


Figure 6: Method mentions across NBER programs. Each line segment connects a program’s 2000–2015 share (left) to its 2016–2024 share (right), colored by field. Steeper upward slopes indicate faster growth. See Table 2 for field definitions.

Which methods have driven the growth? Figure 7 presents a heatmap of the change in method share by program between 2000–2015 and 2016–2024. The answer is unambiguous: DiD accounts for most of the growth across programs. The share of papers mentioning instrumental variables has stayed roughly constant. Regression discontinuity has risen only slightly.⁷ Experiments have

⁷Most of the growth in RD is concentrated in Public Economics, Economics of Aging, and Law and Economics, though the highest levels remain in Economics of Education and Children. Education is perhaps unsurprising given the prominence of test-score cutoffs as a canonical RD design.

risen across programs, with the largest growth in Law and Economics, Productivity, Innovation, and Entrepreneurship, and Economics of Aging.

The pattern for finance and macro is clear: growth in credibility revolution methods has been concentrated in DiD. The credibility revolution in these fields has been, to a first approximation, a difference-in-differences revolution. An important caveat is that keyword matching cannot distinguish between papers using traditional two-way fixed effects specifications and those adopting the robust estimators developed in the recent methodological literature (Callaway, Goodman-Bacon, and Sant’Anna 2024; De Chaisemartin and d’Haultfoeuille 2020; Sun and Abraham 2021). This limits what we can say about the *quality* of DiD adoption, even as we measure its *quantity*. It is possible that late-adopting fields have leapfrogged to better implementations—a silver lining of the adoption lag.

Bartik and shift-share mentions have grown across all programs, but most sharply in International Trade and Investment—where nearly 16 percent of papers now mention them—followed by Development of the American Economy and Labor Studies.⁸ Synthetic control mentions have also grown, with the largest increases in Law and Economics, Health Economics, and Children.

2.4 The dominance of difference-in-differences across fields

How much does this single method account for the overall growth in experimental and quasi-experimental methods? A simple exercise makes the answer concrete. Figure 8 compares method shares with and without DiD.

Panel (a) breaks down the comparison by field. Over the 2016–2024 period, including DiD raises finance’s methods share by 15.7 percentage points—a 53 percent increase—compared to a similar 13.7 percentage point increase for applied micro, which represents only a 29 percent increase because applied micro’s baseline is much higher. For macro/other, the absolute gap is similar (about 12.1 percentage points), and the percentage increase is 38 percent. Finance’s methodological transformation is, to a striking degree, a DiD story.

Panel (b) decomposes the percentage increase by program for 2016–2024. International Finance and Macroeconomics shows the largest increase, followed by Corporate Finance, Development of the American Economy, and Asset Pricing. By contrast, applied micro programs with high overall method shares—such as Development Economics and Education—show relatively small increases from DiD. This reflects their diversified methodological portfolios: these programs rely on a mix of RCTs, RD, and other designs, not just DiD.

Because the composite “DiD” measure flags any mention of either “difference-in-differences” or “event study,” part of the finance increase could in principle come from abnormal-return-style financial event studies that are methodologically distinct from DiD-style designs. Appendix C (Figure 16) decomposes the composite into strict-DiD-only mentions (panel a), any event-study mention (panel b), event-study mentions without strict DiD (panel c), and event-study mentions co-occurring

⁸This likely reflects the use of shift-share instruments in studying the China Shock (Autor, Dorn, and Hanson 2013), historical migration (Boustan 2010), and the long tradition of Bartik instruments in labor studies (Bartik 1991).

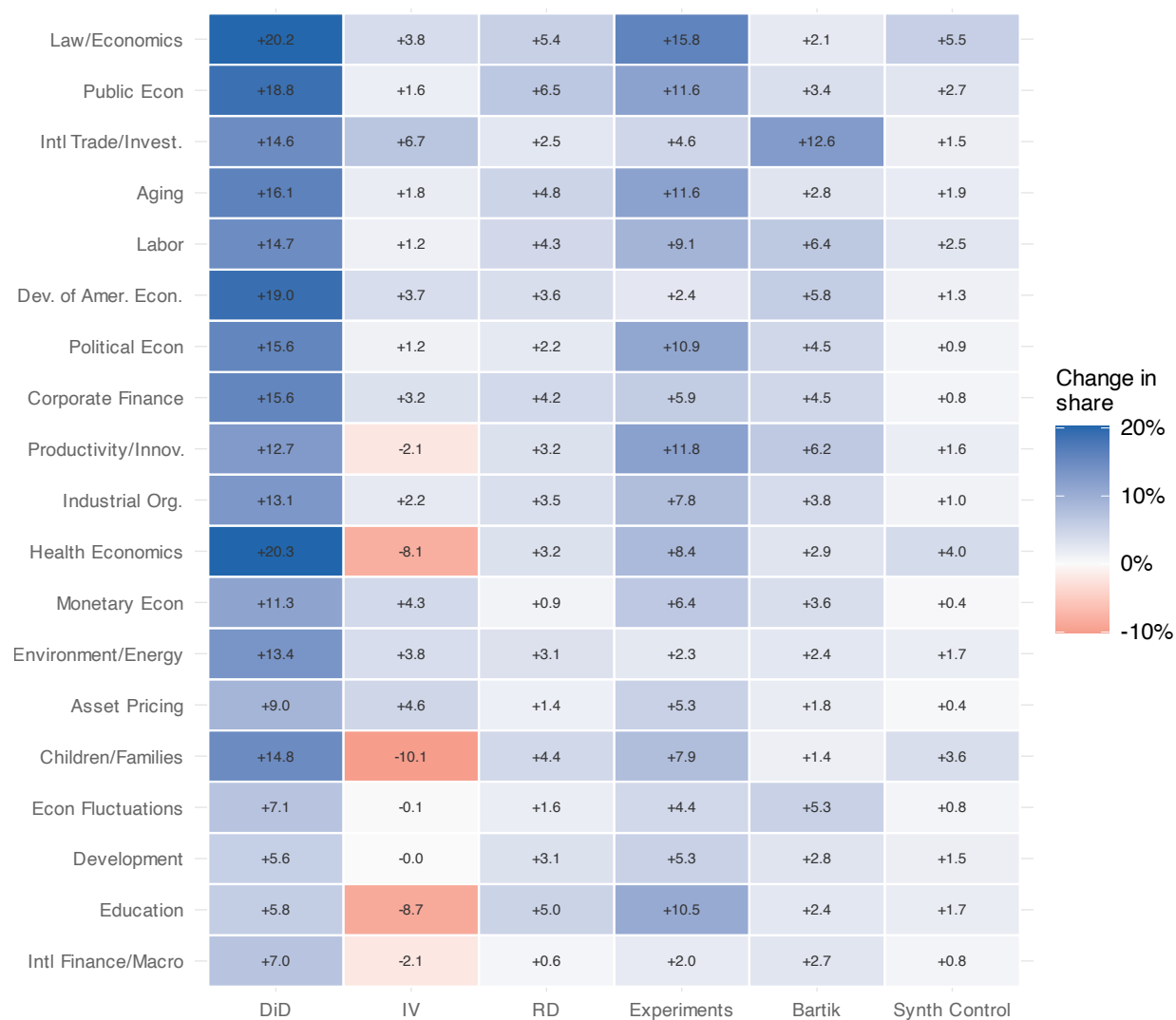


Figure 7: Change in method-specific mentions across NBER programs, 2016–2024 minus 2000–2015. Each cell shows the percentage-point change in share. Blue indicates growth, red indicates decline. See Table 2 for field definitions.

with “abnormal return” as a proxy for financial event studies (panel d). In finance, a substantial share of the event-study-only mentions also contain “abnormal return.” The strict-DiD share, however, also rises markedly in finance over the sample period, so the convergence is not driven entirely by financial event studies—but the headline percentage in panel (a) should be read with the strict-DiD decomposition in mind.

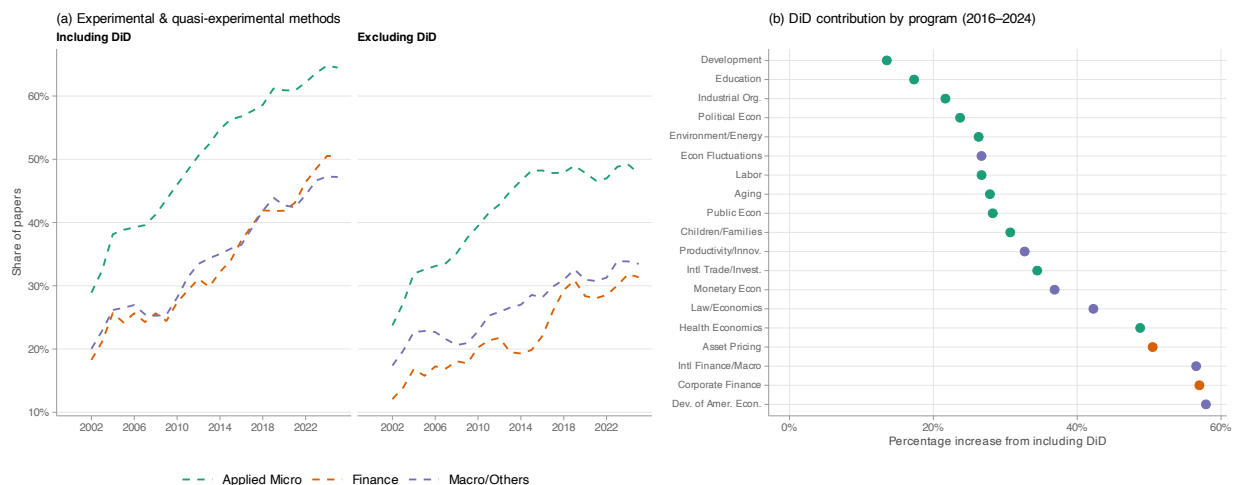


Figure 8: The dominance of difference-in-differences. Panel (a): experimental and quasi-experimental method shares by field, faceted by whether DiD is included or excluded. Panel (b): percentage increase in method share from including DiD, by NBER program (2016–2024). See Table 2 for field definitions.

3 Evidence from Top Journals

The NBER working paper series is a natural laboratory for studying methodological trends, but it has a limitation: NBER affiliates are a selected group, and working papers may differ systematically from published articles. Do the patterns above survive in a different sample? I examine published articles from top economics and finance journals. This analysis serves two purposes: it provides parallel evidence on the NBER trends from a partially overlapping corpus (many NBER working papers eventually appear in these journals, so the two samples are not fully independent), and it gives a first look at whether the publication process amplifies or attenuates the credibility revolution’s reach.

3.1 Overall trends across top journals

Figure 9 compares identification and DiD mentions across journal fields between 2011–2017 and 2018–2024. Applied micro journals lead on both measures in both periods. The two non-applied-micro fields look quite different across the two panels, however: in identification (panel a), macro—driven mainly by AEJ Macroeconomics—runs close to finance in both periods, while in DiD (panel b), finance runs well above macro. The DiD pattern matches the NBER data closely; the identification

pattern is more sensitive to the journal-level field classification. One important caveat: the journal-level field classification for macro is considerably noisier than for the NBER sample. In the NBER data, macro is defined by program affiliation; in the journal data, macro is represented primarily by AEJ Macroeconomics plus JEL-classified papers from the AER. QJE and JPE do not report JEL codes, so macro papers in those journals default to “General Econ” and are excluded from the field comparison. The journal-level macro trends should therefore be interpreted with more caution than the finance or applied micro trends, which benefit from dedicated field journals. Appendix J shows that the cross-field patterns hold when restricting to field-specific journals only.

The DiD pattern (panel b) is notable: applied micro journals show the steepest growth in DiD mentions (from 27% to 40%), while finance shows steady but more modest growth (from 26% to 32%), confirming the NBER finding that the credibility revolution in finance remains substantially behind applied micro. Full time series for all four method categories, including instrumental variables and structural models, are reported in Appendix I.

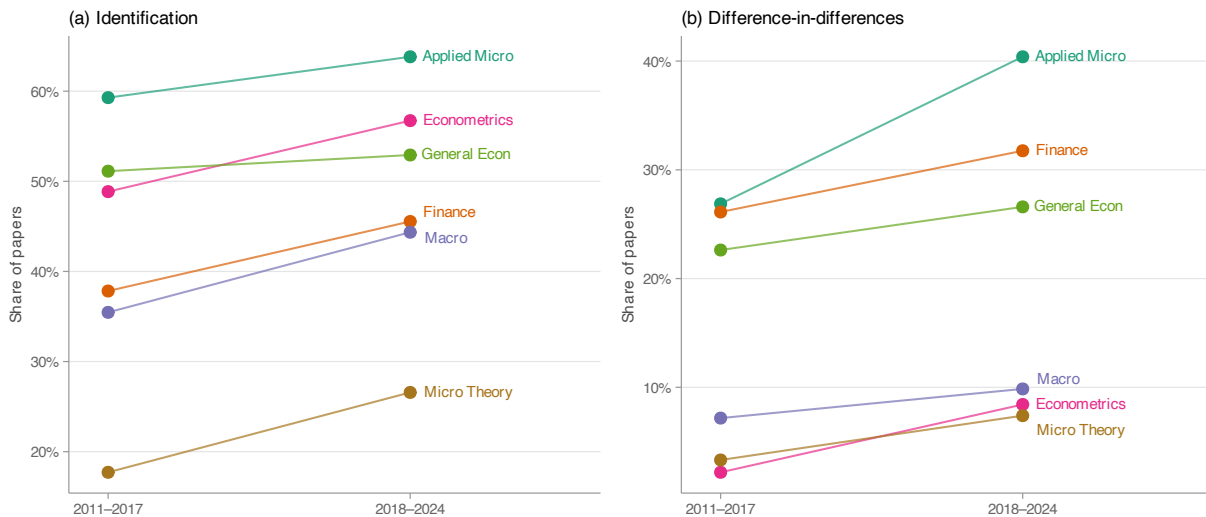


Figure 9: Method mentions across fields in top journals: 2011–2017 vs. 2018–2024. Each line connects a field’s early-period share (left) to its late-period share (right). Panel (a) identification, (b) difference-in-differences. See text for journal-to-field mapping.

3.2 Comparison across individual journals

Figure 10 compares mention rates across individual journals. AEJ Applied Economics and AEJ Economic Policy show the highest rates of credibility revolution methods—unsurprising given their explicit focus on applied empirical work. Among the general-interest journals, AER and QJE show higher rates than JPE, reflecting differences in paper composition. The finance journals show moderate adoption of DiD and identification language but lower rates of RD and experimental methods—echoing the NBER findings at the journal level.

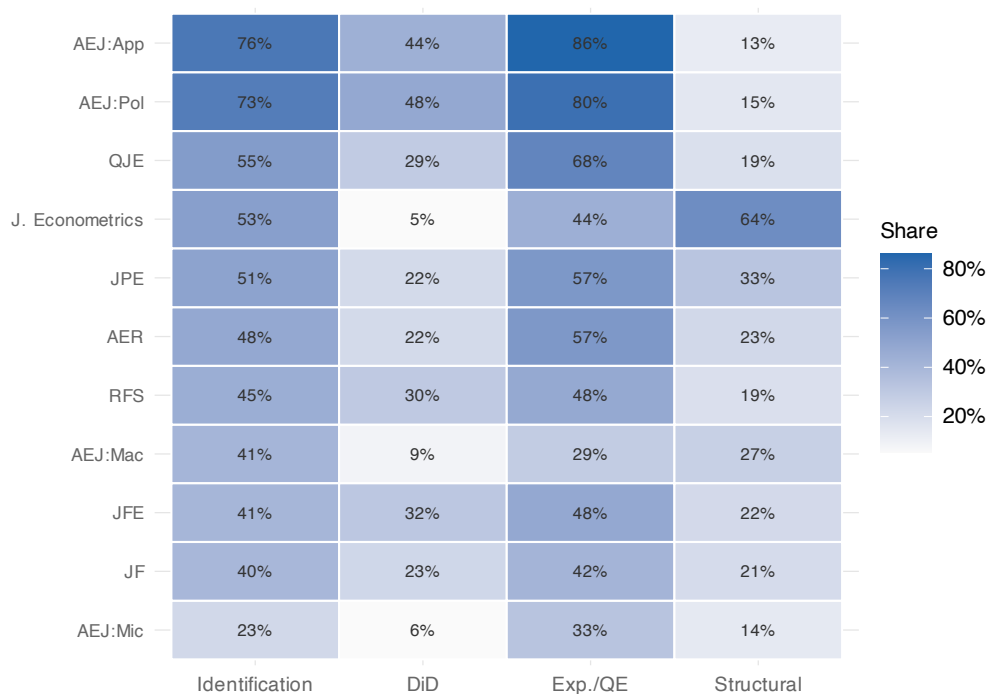


Figure 10: Heatmap of method mentions across individual journals (2011–2024). Color intensity reflects the share of papers mentioning each method category. Journals ordered by identification share.

3.3 NBER working papers vs. published articles

Could the NBER trends be artifacts of the working paper selection process? Figure 11 overlays the NBER and journal time series for key methods, matching by field. The trends are strikingly similar. Published articles show slightly higher rates of most credibility revolution methods, which is at least suggestive of a selection effect where papers using transparent research designs are more likely to clear the bar at top journals. However, the two samples are not fully independent, since many of the top-journal articles circulated first as NBER working papers. A cleaner test would link NBER working papers to their eventual published versions; the parallel trends here are best read as evidence that the patterns in Section 2 are not an artifact of NBER-specific selection, rather than as proof of publication-stage selection.

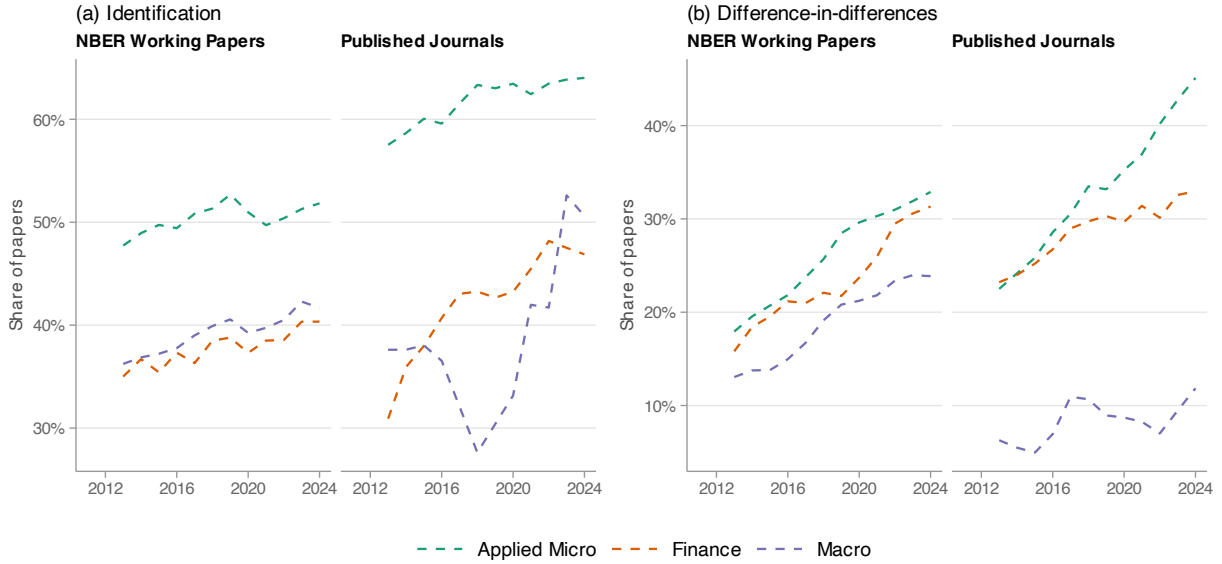


Figure 11: NBER working papers vs. published journal articles: time series by field (2011–2024). Each panel compares NBER working papers (left facet) with published journals (right facet) for identification (left) and difference-in-differences (right).

4 The Topic Mix in the *Journal of Econometrics*

Having documented uneven diffusion of credibility revolution methods across applied fields, I now turn to a narrower question: what is the topic mix in the one methods journal in my corpus, the *Journal of Econometrics*? The credibility revolution depends on tools developed by econometricians—DiD estimators, IV techniques, RD designs, synthetic control methods—all of which required substantial theoretical development. Looking at where these tools show up in the methods literature gives one slice of how econometric theory and applied practice relate. The scope is genuinely limited: I have full-text coverage of the *Journal of Econometrics*, but other major methods outlets—*Econometrica*, the *Review of Economic Studies*, the *Journal of the American Statistical Association*, and *Econometric Theory*—are not in my corpus, and many of the influential credibility-revolution methods papers appeared in these other outlets. The patterns here describe *Journal of Econometrics* content, not the universe of econometric theory.

To make this concrete, I compare term prevalence in the *Journal of Econometrics* (2,159 papers, 2011–2024) with that in the applied economics and finance journals in the corpus. I begin with a natural question: of the methods that applied researchers care about, what share of *Journal of Econometrics* papers touch on them? Panel (a) of Figure 12 shows the answer. Most credibility revolution methods—DiD, event studies, RD, RCTs, administrative data, synthetic control, Bartik instruments, binscatter, and heterogeneous treatment effects—appear far less frequently in the *Journal of Econometrics* than in applied journals. DiD appears in approximately 21% of applied journal papers but under 5% of *Journal of Econometrics* papers; event studies show a similar gap. The exceptions are identification language and instrumental variables, where the *Journal of Econo-*

metrics matches or exceeds applied journals—reflecting the theoretical literature on identification and IV estimation that is a core focus of the journal.

This raises a natural follow-up: if the *Journal of Econometrics* is not primarily publishing on applied methods, what *does* it publish? Panel (b) takes a data-driven approach. I start by constructing keyword lists for twenty candidate topic areas in econometric theory, drawn from the major sections of standard graduate econometrics textbooks and the *Journal of Econometrics*’ own subject classifications: nonparametric estimation, semiparametric estimation, time series models (VAR, GARCH, cointegration, unit root), Bayesian methods, bootstrap and resampling, machine learning, regularization, panel data methods, limited dependent variable models, quantile regression, kernel methods, forecasting, robust inference, weak identification, high-dimensional methods, simulation/Monte Carlo, asymptotic theory, treatment effects, and structural/GMM/MLE methods.⁹ I then compute the prevalence of each category in the *Journal of Econometrics* and in applied journals, rank by *Journal of Econometrics* prevalence, and show the top fifteen alongside the corresponding rates in applied journals. Composite categories and narrow variants are excluded to avoid double-counting.¹⁰ Asymptotic theory and Monte Carlo simulation top the list—appearing in 87% and 68% of papers respectively. These are the standard toolkit for deriving and validating estimators, so their dominance is unsurprising and partly mechanical: any paper that establishes large-sample properties or runs simulation experiments will trigger these flags. The simulation/Monte Carlo category in particular also overlaps with the Bayesian category through MCMC, Markov chain Monte Carlo, and Gibbs sampling triggers, so the 68% should be read as a broad “simulation or computational posterior inference” measure rather than a narrow “simulation used to validate estimators” measure. The more informative contrasts involve substantive methods: nonparametric estimation (64%), structural/GMM/MLE methods (62%), time series models (58%), and Bayesian methods all appear at much higher rates in the *Journal of Econometrics* than in applied journals. These are the estimation and inference techniques that form a sizable share of *Journal of Econometrics* content—important in their own right, but distant from the day-to-day practice of most applied economists. The gap between the two panels illustrates how the *Journal of Econometrics* and applied journals occupy largely separate methodological spaces, at least within the corpus I observe.

Figure 13 makes the full picture concrete through a heatmap of key terms across all journals. The *Journal of Econometrics* has a strikingly different methodological profile from every other journal in the sample—high rates of asymptotic theory and simulation (reflecting the standard tools for proving and testing estimators), along with nonparametric estimation, time series models, and Bayesian methods, but low rates of DiD and event studies. Appendix L shows that credibility revolution methods have grown within the *Journal of Econometrics* over the sample period, suggesting the gap may be narrowing.

Three caveats are important.

⁹The full list of trigger phrases for each category appears in the Appendix. The core applied categories follow Currie, Kleven, and Zwiars (2020b); the econometrics categories are new to this paper.

¹⁰I exclude composites like “all experiments” (which unions DiD, RCTs, etc.) and narrow variants like “synth DiD” (subsumed by “synth control”).

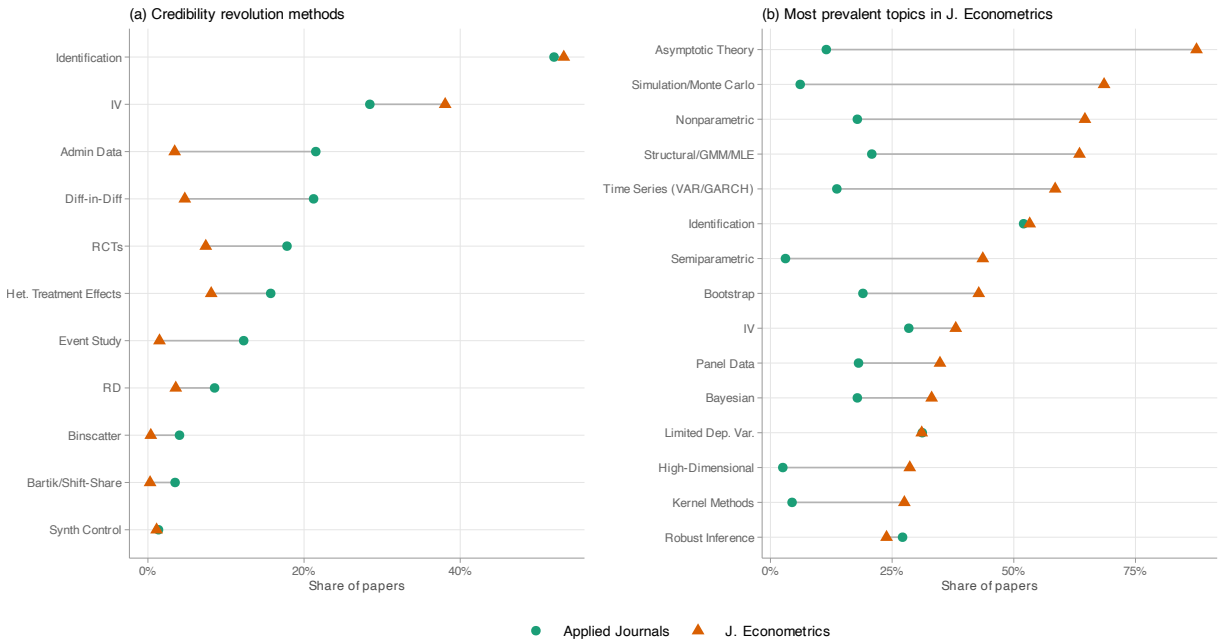


Figure 12: Comparison of term prevalence: *Journal of Econometrics* vs. applied economics journals (2011–2024). Panel (a) shows credibility revolution methods; panel (b) shows the fifteen most prevalent topics in the *Journal of Econometrics*, ranked by share.

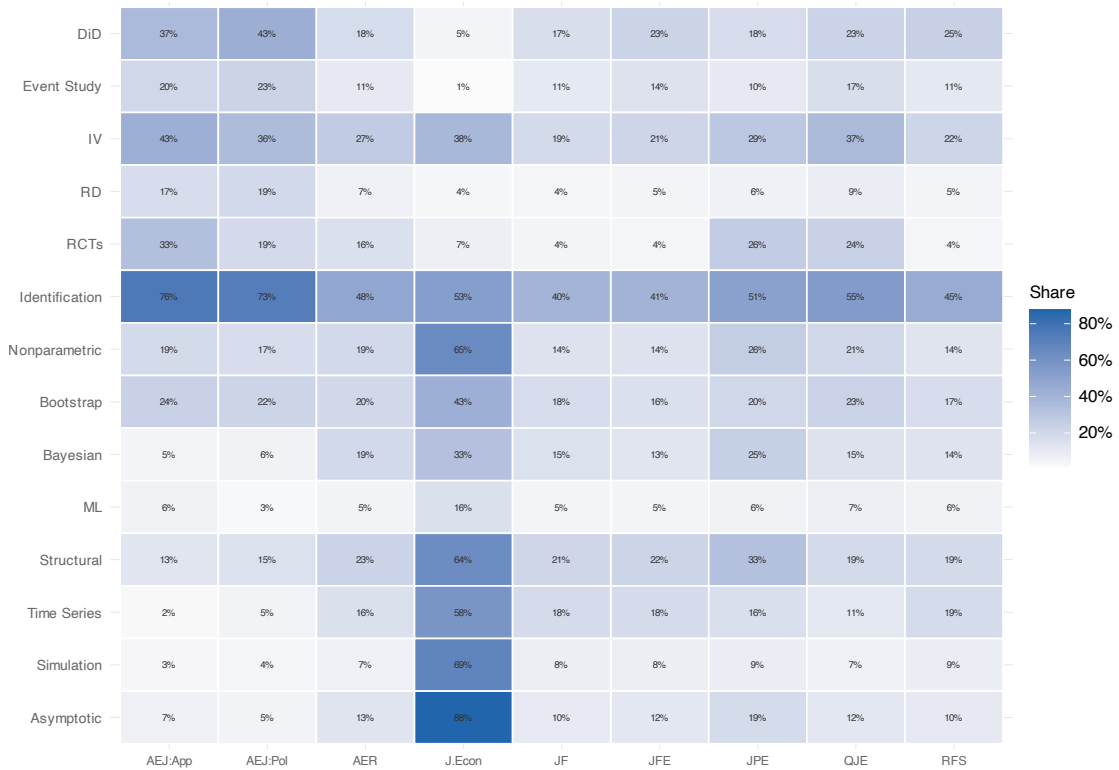


Figure 13: Heatmap of method term prevalence across journals (2011–2024). Color intensity reflects the share of papers mentioning each term.

First, the econometrics articles (including those published in the *Journal of Econometrics*) have been influential when it engages directly with the credibility revolution’s tools. For example, the literatures on heterogeneous treatment effects in DiD (De Chaisemartin and d’Haultfoeuille 2020; Goodman-Bacon 2021), parallel-trends sensitivity and pre-testing (Roth 2022; Rambachan and Roth 2023), continuous-treatment DiD (Callaway, Goodman-Bacon, and Sant’Anna 2024), and machine learning for causal inference have reshaped applied practice. However, contributions of this type that do appear in the journal represent a relatively small slice of what it publishes. Panel (b) of Figure 12 shows that the bulk of the journal focuses on nonparametric estimation, time series models, and Bayesian methods—problems important in their own right but distant from the day-to-day practice of applied economics. This suggests an opportunity: the journal’s outsized impact on applied work, when it chooses to engage with applied tools, makes it a natural venue for bridging the gap documented here.

Second, the gap could reflect productive intellectual specialization rather than misalignment, with applied and theoretical work advancing on parallel tracks that occasionally intersect. Methods journals may be developing tools that will eventually diffuse to practice—as happened with the DiD robustness literature, which moved from econometric theory to widespread applied adoption in under five years.

Third, the cross-field differences documented in Sections 2 and 3 should not be read as implying that all fields *should* converge to the applied micro toolkit. Many questions in macroeconomics are fundamentally about general equilibrium, and the applied micro toolkit—built around partial equilibrium and local treatment effects—may not be the right tool for every setting. The same is true in asset pricing, where the object of interest is often an equilibrium price rather than a treatment effect. The more relevant distinction is between fields where quasi-experimental methods are feasible but underused—corporate finance, for example, has abundant natural experiments—and fields where the questions themselves call for different approaches. Nakamura and Steinsson (2018) offer a thoughtful example of how credibility revolution thinking can be adapted to macroeconomic settings without simply importing the applied micro playbook.

Why does this gap matter? Because the rare instances where the two literatures *do* intersect have been extraordinarily productive. The DiD robustness literature—Callaway, Goodman-Bacon, and Sant’Anna (2024), Goodman-Bacon (2021), De Chaisemartin and d’Haultfoeuille (2020), Sun and Abraham (2021)—moved from econometric methodology to widespread applied adoption in under five years, fundamentally changing how researchers implement one of the most common research designs. These examples demonstrate that the gap is not inherent: when econometric methodologists engage with the tools applied researchers actually use, the payoff for both sides is large. The gap documented here thus represents an opportunity, not just a description.

5 Conclusion

The credibility revolution has continued to advance, but the picture is one of uneven progress rather than uniform transformation. Three patterns stand out.

First, credibility revolution methods remain most prevalent in applied microeconomics. Finance and macro/other have more mentions of causal inference methods since the early 2000s, but they differ substantially from applied micro on most measures—with current levels comparable to where applied micro was more than a decade ago. Within finance, growth is concentrated in corporate finance; within macro/other, there is wide variation across programs. Whether these gaps reflect a lag that will close over time or different long-run equilibria—driven by the nature of the research questions and the available data—is an important open question.

Second, outside applied micro, the credibility revolution has been—to a first approximation—a difference-in-differences revolution. Over the 2016–2024 period, including DiD raises the finance methods share by roughly 53 percent versus 29 percent for applied micro. Other quasi-experimental approaches—regression discontinuity, RCTs, instrumental variables—have seen far less growth in finance and macro. This concentration on a single method is noteworthy given the recent econometrics literature highlighting important sensitivities in DiD designs—most prominently, the work of Callaway, Goodman-Bacon, and Sant’Anna (2024) and De Chaisemartin and d’Haultfoeuille (2020) showing that conventional two-way fixed effects estimators can be severely biased under treatment effect heterogeneity, and the growing literature on robust alternatives (Roth 2022; Roth and Sant’Anna 2023; Rambachan and Roth 2023; Chaisemartin et al. 2022; Sun and Abraham 2021). The rapid diffusion of these methodological refinements reflects the productive interplay between econometric theory and applied practice—and suggests that the concentration on DiD may be less concerning if practitioners are adopting improved estimators alongside the research design itself. Notably, synthetic control methods—which share many properties with DiD—have not experienced comparable growth and may even be declining.

Third, within the methods journal I have full-text coverage of, the topic mix is quite different from applied practice. The *Journal of Econometrics* is dominated by nonparametric estimation, time series models, and Bayesian methods—alongside the asymptotic theory and simulation tools used to develop and validate estimators. The applied literature runs on DiD, identification strategies, and administrative data. These two literatures occupy largely separate spaces within this corpus. This is a narrower claim than a universal econometrics-versus-applied disconnect: many influential methods papers are published in *Econometrica*, the *Review of Economic Studies*, the *Journal of the American Statistical Association*, and field journals that are not in the corpus here.

These patterns hold across data sources: NBER working paper trends are confirmed by published articles from eleven top journals, and LLM-based classification validates the keyword approach at 72–92 percent agreement rates for most method categories, though agreement is lower for identification strategy and structural models (Appendix A).

Looking ahead, the dominance of DiD raises a question about the trajectory of the credibility revolution. The revolution’s early promise was methodological pluralism—a toolkit of transparent research designs, each suited to different empirical settings. The data show that this pluralism has been more fully realized in applied micro than elsewhere. As finance and macroeconomics continue to adopt credible methods, there is value in diversifying beyond DiD, both to strengthen the robustness of individual studies and to expand the set of questions these fields can credibly

address.

What should researchers in finance and macro take from these patterns? The answer depends on the setting. In corporate finance, where natural experiments and policy variation are often available, the gap likely reflects unrealized potential: there are many settings where DiD, RD, or IV could be applied but have not yet been. In asset pricing and general equilibrium macro, the gap may partly reflect the nature of the questions—structural models are sometimes the right tool, not a methodological shortcoming. The useful question is not “why doesn’t macro look like applied micro?” but rather “are there settings in these fields where credible research designs would sharpen inference?”

One limitation of this analysis is that keyword mentions measure the *diffusion of methodological language* but not the *quality of adoption* or *influence* of methods. Validation against LLM classification (Appendix A) shows two distinct properties of the keyword measure. *Accuracy*—the share of papers where the keyword flag agrees with the LLM, including joint negatives—exceeds 90 percent for regression discontinuity and lab experiments and runs 74–86 percent for DiD, IV, RCT, and event studies. *Precision*—the share of keyword-flagged papers the LLM confirms as actually using the method—is much lower, generally below 50 percent across these categories. The high accuracy figures are partly driven by joint negatives (papers that neither measure flags), while low precision reflects that the keyword search captures mentions in literature reviews, robustness checks, and discussion rather than primary research designs. Cross-field comparisons should therefore be interpreted with caution for categories where precision is lowest, as some of the measured gap may reflect differences in vocabulary rather than uptake. A complementary approach would track citations to foundational credibility revolution papers—Angrist and Krueger (1991), Angrist and Pischke (2009), Imbens and Lemieux (2008)—across fields. If finance and macro cite these works at comparable rates but describe methods differently, the measured gap would partly reflect writing conventions rather than substantive methodological differences. If citation rates also differ, this would reinforce the keyword evidence. This is a natural direction for future work.

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Table 4: Search Categories and Trigger Phrases. Unless noted otherwise, the outcome is the fraction of papers with at least one phrase match. “Figure” and “Table” categories use average word count per paper.

Category	Trigger Phrases	Case Sens.	Wildcard	Cond. data
Administrative Data	'administrative data', 'admin data', 'administrative-data', 'admin-data', 'administrative record', 'admin record', 'administrative regist', 'admin regist', 'register data', 'registry data'	No	Yes	Yes
Bartik / Shift-share	'bartik', 'shift-share', 'shift share', 'shiftshare' (case-insensitive regex match)	No	Yes	No
Big Data	'big data', 'big-data'	No	Yes	Yes
Binscatter	'binscatter', 'bin scatter', 'binned scatter'	No	Yes	No
Bunching	'bunching'	No	Yes	No
Clustering	'cluster'	No	Yes	Yes
Confidence Interval	'confidence interval'	No	Yes	Yes
Data	'data'	No	Yes	No
Difference-in-Differences	'Difference in Diff', 'Difference in diff', 'difference in diff', 'Difference-in-Diff', 'Difference-in-diff', 'difference-in-diff', 'Differences in Diff', 'Differences in diff', 'differences in diff', 'Differences-in-Diff', 'Differences-in-diff', 'differences-in-diff', 'diff-in-diff', 'd-in-d', 'DiD'	Yes	Yes	No
Event Study	'event stud' 'event-stud'	No	Yes	No
External Validity	'external validity', 'external-validity', 'externally valid', 'externally-valid'	No	Yes	No
Figure	'graph', 'figure', 'plot', 'chart'	No	Yes	No
Fixed Effects	'FE', 'Fixed Effect', 'Fixed effect', 'fixed effect', 'Fixed Effects', 'Fixed effects', 'fixed effects', 'Fixed-Effect', 'Fixed-effect', 'fixed-effect', 'Fixed-Effects', 'Fixed-effects', 'fixed-effects'	Yes	No	Yes

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Table 4 – *Continued from previous page*

Category	Trigger Phrases	Case Sens.	Wildcard	Cond. data
Functional Forms	'CES', 'constant elasticity of substitution', 'Constant Elasticity of Substitution', 'Constant elasticity of substitution', 'Cobb-Douglas', 'Cobb Douglas', 'Stone Geary', 'Stone-Geary', 'CRRA', 'coefficient of relative risk-aversion', 'coefficient of relative risk aversion', 'Coefficient of relative risk-aversion', 'Coefficient of relative risk aversion', 'Coefficient of Relative Risk-Aversion', 'Coefficient of Relative Risk Aversion', 'CARA', 'constant absolute risk aversion', 'constant absolute risk-aversion', 'Constant absolute risk aversion', 'Constant absolute risk-aversion', 'Constant Absolute Risk Aversion', 'Constant Absolute Risk-Aversion', 'translog', 'Translog'	Yes	No	No
General Equilibrium	'general equilibr', 'general-equilibr'	No	Yes	No

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Table 4 – *Continued from previous page*

Category	Trigger Phrases	Case Sens.	Wildcard	Cond. data
Identification	Sentence structure: search for sentences that have the term 'identif' in combination with any of the terms: 'effect', 'response', 'impact', 'elasticit', 'parameter', or 'coefficient' with maximum two words in between. Note that even though the search includes wildcards at the end, we exclude any match with the word 'effective'. Also search for these terms: 'causal identification', 'causally identified', 'identification strategy', 'identification approach', 'identification assumption', 'identifying assumption', 'identifying variation', 'empirical identification', 'over identified', 'over-identified', 'under identified', 'under-identified', 'identification properties', 'identification test', 'identification problem', 'identification issue', 'problem with identification', 'problems with identification', 'issue with identification', 'issues with identification', 'problem identifying', 'problems identifying', 'issue identifying', 'issues identifying', 'threat to identification', 'threats to identification', 'threat for identification', 'threats for identification', 'over identifying', 'over-identifying', 'under identifying', 'under-identifying', 'partial identification', 'partially identified', 'non-parametric identification', 'nonparametric identification', 'non parametric identification', 'non-parametrically identified', 'nonparametrically identified', 'non parametrically identified', 'identification condition', 'identifying condition', 'condition for identification', 'conditions for identification', 'condition for identifying', 'conditions for identifying', 'point identification', 'point-identification', 'point identified', 'point-identified', 'point identifying', 'point-identifying', 'set identification', 'set-identification', 'set identified', 'set-identified', 'set identifying', 'set-identifying', 'identification analysis', 'weak identification', 'identification result', 'identification argument', 'identification framework', 'identification scheme'	No	Yes	No
Internet Data	'internet data', 'internet-data', 'web data', 'web-data', 'scraped data', 'scraped-data', 'scrape data', 'scraping data', 'search data', 'search-data', 'google data', 'google-data', 'social media data', 'google trend', 'google-trend', 'google search', 'google-search', 'google ngram', 'google n-gram', 'google books ngram', 'google books n-gram'	No	Yes	Yes

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Table 4 – *Continued from previous page*

Category	Trigger Phrases	Case Sens.	Wildcard	Cond. data
Instrumental Variables	'Instrumental Variable', 'Instrumental variable', 'instrumental variable', 'Instrumental-Variable', 'Instrumental-variable', 'instrumental-variable', 'Two Stage Least Squares', 'Two stage least squares', 'two stage least squares', '2SLS', 'TSLS', 'valid instrument', 'exogenous instrument', 'IV Estim', 'IV estimat', 'IV-estimat', 'IV Specification', 'IV specification', 'IV-specification', 'IV Regression', 'IV regression', 'IV-regression', 'IV Strateg', 'IV strateg', 'IV-strateg', 'we instrument', 'I instrument', 'paper instruments', 'exclusion restriction', 'weak first stage', 'simulated instrument'	Yes	Yes	Yes
Lab Experiments	'Laboratory Experiment', 'Laboratory experiment', 'laboratory experiment', 'Lab Experiment', 'Lab experiment', 'lab experiment', 'Dictator Game', 'Dictator game', 'dictator game', 'Ultimatum Game', 'Ultimatum game', 'ultimatum game', 'Trust Game', 'Trust game', 'trust game', 'Public Good Game', 'Public good game', 'public good game', 'Public Goods Game', 'Public goods game', 'public goods game', 'Z-tree', 'zTree', 'ORSEE', 'show-up fee', 'laboratory participant', 'lab participant'	Yes	Yes	No
Machine Learning	'machine learning', 'lasso', 'random forest'	No	Yes	No
Matching	'propensity score', 'propensity score matching', 'propensity-score matching', 'matching estimat', 'nearest neighbor matching', 'nearest-neighbor matching', 'nearest neighbour matching', 'nearest-neighbour matching', 'caliper matching', 'stratification matching', 'exact matching', 'one to one matching', 'one-to-one matching', 'kernel matching', 'inverse probability matching', 'inverse-probability matching'	No	Yes	Yes
Mechanisms	'mechanism'	No	Yes	No
Omitted Variables	'omitted variable'	No	Yes	Yes
Preanalysis Plan	'pre-analysis plan', 'pre analysis plan', 'preanalysis plan'	No	Yes	No
Precisely Estimated	'precisely estimated', 'precisely-estimated'	No	Yes	No

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Table 4 – *Continued from previous page*

Category		Trigger Phrases	Case Sens.	Wildcard	Cond. data
Precisely Estimated Zero		'precisely estimated zero', 'precisely-estimated zero'	No	Yes	No
Proprietary Data		'proprietary data', 'confidential data', 'nonpublic data', 'non-public data', 'proprietary-data', 'confidential-data', 'nonpublic-data', 'non-public-data'	No	Yes	Yes
Quasi- and Natural Experiments		'quasi experiment', 'quasi-experiment', 'quasiexperiment', 'natural experiment', 'natural-experiment'	No	Yes	No
RCTs		'Randomized Controlled Trial', 'Randomized controlled trial', 'randomized controlled trial', 'Randomized Control Trial', 'Randomized control trial', 'randomized control trial', 'Randomized Field Experiment', 'Randomized field experiment', 'randomized field experiment', 'Randomized Controlled Experiment', 'Randomized controlled experiment', 'randomized controlled experiment', 'Randomised Controlled Trial', 'Randomised controlled trial', 'randomised controlled trial', 'Randomised Control Trial', 'Randomised control trial', 'randomised control trial', 'Randomised Field Experiment', 'Randomised field experiment', 'randomised field experiment', 'Randomised Controlled Experiment', 'Randomised controlled experiment', 'randomised controlled experiment', 'Social Experiment', 'Social experiment', 'social experiment', 'RCT', 'randomized experiment', 'randomised experiment', 'randomized evaluation', 'randomised evaluation', 'randomized trial', 'randomised trial', 'randomized intervention', 'randomised intervention', 'randomized design', 'randomised design', 'field experiment'	Yes	Yes	No
Regression discontinuity	Dis-	'Regression Discontinuit', 'Regression discontinuit', 'regression discontinuit', 'Regression-discontinuity', 'regression-discontinuity', 'Regression Kink', 'Regression kink', 'regression kink', 'RD Design', 'RD design', 'RD-design', 'RD Estim', 'RD estimat', 'RD-estimat', 'RD Model', 'RD model', 'RD-model', 'RD Regression', 'RD regression', 'RD-regression', 'RD Coefficient', 'RD coefficient', 'RD-coefficient', 'RK Design', 'RK design', 'RK-Design', 'RK-design', 'RKD', 'RDD'	Yes	Yes	No
Reverse Causation		'reverse causa', 'reverse-causa'	No	Yes	Yes

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Table 4 – *Continued from previous page*

Category	Trigger Phrases	Case Sens.	Wildcard	Cond. data
Selection	'selection'	No	Yes	Yes
Simultaneity	'simultaneity'	No	Yes	Yes
Structural Model	Sentence structure: we search for instances where, within two full stops, the term 'structural' is mentioned in combination with either 'model', 'specification', 'estimate', or 'parameter'. Also search for these terms: 'Structural Model', 'Structural model', 'structural model', 'Method of Moments', 'Method of moments', 'method of moments', 'Method-of-Moments', 'Method-of-moments', 'method-of-moments', 'Berry, Levinsohn, Pakes', 'Berry, Levinsohn and Pakes', 'Berry, Levinsohn, and Pakes', 'BLP', 'Structural General Equilibrium Model', 'Structural general equilibrium model', 'structural general equilibrium model', 'GMM', 'Maximum Likelihood Estimato', 'Maximum likelihood estimato', 'maximum likelihood estimato', 'Maximum-Likelihood Estimato', 'Maximum-likelihood estimato', 'maximum-likelihood estimato', 'MLE'	Yes	Yes	No
Survey Data	Sentence structure: we search for instances where the term 'survey' and 'data' are mentioned within two full stops.	No	Yes	Yes
Synthetic Control	'synthetic control'	No	Yes	Yes
Table	'table'	No	Yes	No
Text Analysis	'natural language processing', 'text analys', 'computational linguistics', 'speech processing', 'n-gram', 'ngram', 'n gram', 'textual analys', 'language processing', 'language analys', 'text data', 'text mining', 'mining text', 'text regression', 'tokeniz'	No	Yes	No
<i>Econometrics categories (Section 4)</i>				
Asymptotic Theory	'asymptot', 'large sample', 'convergence rate', 'consistency', 'limiting distribut'	No	Yes	No
Bayesian	'Bayesian', 'posterior distribut', 'prior distribut', 'Markov chain Monte Carlo', 'MCMC'	Mixed	Yes	No
Bootstrap	'bootstrap', 'resampl'	No	Yes	No

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Table 4 – *Continued from previous page*

Category	Trigger Phrases	Case Sens.	Wildcard	Cond. data
Forecasting	'forecast accuracy', 'forecast error', 'forecast evaluation', 'forecast combination', 'out of sample forecast', 'predictive regression'	No	Yes	No
High-Dimensional	'high-dimensional', 'high dimensional', 'many regressors', 'many covariates', 'variable selection', 'sparsity', 'sparse model'	No	Yes	No
Kernel Methods	'kernel estimat', 'kernel density', 'kernel regression', 'bandwidth selection', 'kernel smooth'	No	Yes	No
Limited Dep. Var.	'Tobit', 'probit', 'logit', 'censored regression', 'truncated regression', 'sample selection model', 'Heckman', 'discrete choice'	Mixed	Yes	No
Nonparametric	'nonparametric', 'non-parametric', 'non parametric'	No	Yes	No
Panel Data	'panel data', 'longitudinal data', 'fixed effect', 'random effect', 'within estimat', 'between estimat'	No	Yes	No
Quantile Regression	'quantile regression', 'quantile treatment'	No	Yes	No
Regularization	'LASSO', 'ridge regression', 'elastic net', 'regulariz', 'penalized regression', 'shrinkage estimat'	Mixed	Yes	No
Robust Inference	'heteroskedasticity robust', 'HAC', 'Newey-West', 'robust standard error', 'cluster robust', 'wild bootstrap'	Mixed	Yes	No
Semiparametric	'semiparametric', 'semi-parametric', 'semi parametric'	No	Yes	No
Simulation/Monte Carlo	'Monte Carlo', 'MCMC', 'Markov chain Monte', 'Gibbs sampl'	Mixed	Yes	No
Time Series (VAR/GARCH)	'VAR', 'vector autoregress', 'ARMA', 'ARIMA', 'unit root', 'cointegrat', 'GARCH', 'ARCH', 'stationarity', 'impulse response'	Mixed	Yes	No
Treatment Effects	'treatment effect', 'average treatment', 'causal effect'	No	Yes	No
Weak Identification	'weak instrument', 'weak identification', 'Anderson-Rubin', 'Stock-Yogo'	Mixed	Yes	No

A LLM Validation of Keyword Matching

The analysis throughout this paper relies on keyword matching—a simple, scalable approach with obvious limitations. Keywords may produce false positives (papers that mention a method without actually using it) or false negatives (papers that use a method but describe it in non-standard language). Recent work by Garg and Fetzer (2025) uses LLMs for paper classification, while Brodeur, Cook, and Heyes (2024) provide hand-coded benchmarks, raising the question of whether more sophisticated approaches would yield different results.

To assess reliability, I classify a stratified sample of approximately 750 papers using two independent LLMs: Claude Haiku 4.5 (Anthropic 2025) and Qwen3.6-35B-A3B-FP8, both run at temperature 0. For each paper, I provide the first 1,500 words and ask the model to identify which methods are actually *used*, as opposed to merely mentioned. I then compare the LLM classification with the keyword flags. The accuracy, precision, recall, F1, and κ statistics in Tables 5 and 6 use the Qwen labels; the two models produce nearly identical method-specific positive rates for most categories (within one to two percentage points), diverging mainly for event studies and administrative data (Table 7), so the headline conclusions do not depend on which model is treated as ground truth. Because the LLMs see only the first 1,500 words of each paper, these statistics are validation against a truncated-paper LLM benchmark rather than full-paper ground truth; methods introduced in later empirical-strategy, data, robustness, or appendix sections may be flagged by the (full-text) keyword search but not by the (truncated-text) LLM, contributing to apparent “keyword false positives.”

The sample is stratified to ensure adequate representation of each method. I select at least 50 papers flagged by keywords for each of the nine method categories, at least 100 papers with no method flags (to detect false negatives), and fill the remainder with a stratified random sample split equally between NBER and journal papers. Because the sample overrepresents rare methods relative to the population, the agreement rates reported here should be interpreted as method-specific accuracy rather than overall population accuracy. If 80 percent of papers do not mention any method, a high false-negative rate on a rare method could affect population-level trend estimates while barely showing up in these agreement rates. The main text addresses this concern in part by showing that NBER and journal trends are consistent, and by decomposing composite measures (e.g., DiD vs. event studies) to isolate potential sources of terminological conflation.

Table 5 presents the results. For most categories, keyword matching achieves 72–92 percent agreement with LLM classification. Agreement is highest for regression discontinuity and lab experiments—where the terminology is distinctive and unlikely to appear outside its intended context. Agreement is lower for identification (where keyword patterns may flag theoretical discussions) and structural models (where terms like GMM and MLE appear in many contexts beyond structural estimation).

Agreement rates also vary by field (Table 6). Agreement tends to be higher in finance and macro, where quasi-experimental methods are rarer and keyword mentions more likely to reflect actual use. Applied micro shows slightly lower agreement, particularly for identification strategy, reflecting the

Table 5: Keyword vs. LLM Classification Agreement

Method	N	Accuracy	Precision	Recall	F1	κ
DiD	750	84.1%	42.6%	89.0%	57.7%	0.493
Event Study	750	86.0%	31.4%	79.6%	45.0%	0.387
IV	750	74.3%	22.5%	100.0%	36.7%	0.279
RD	750	91.5%	36.6%	100.0%	53.6%	0.500
RCT	750	85.5%	37.7%	85.7%	52.4%	0.453
Lab Experiment	750	91.6%	26.3%	84.0%	40.0%	0.368
Identification Strategy	750	57.9%	31.3%	77.4%	44.6%	0.195
Structural Model	750	66.5%	46.5%	36.1%	40.7%	0.179
Administrative Data	750	72.4%	76.5%	43.8%	55.7%	0.377

Notes: Keyword-based classification treated as positive when any pattern matches. LLM classification uses Qwen3.6-35B-A3B-FP8 with temperature 0. Precision and recall measured with keyword as the classifier and LLM as ground truth. Sample of 750 papers stratified by source, field, and method.

higher base rate and more varied terminology in that field. The “Overall” column in Table 6 is the paper-field-observation rate (each paper contributes one observation per field it is listed in), so it does not exactly match the paper-level accuracy in Table 5; the two move together but can diverge for methods whose accuracy varies sharply by field.

Table 6: Keyword vs. LLM Classification Agreement by Field

Method	Overall	Applied Micro	Finance	Macro/Others
DiD	84.1%	81.3%	85.3%	89.4%
Event Study	86.0%	84.6%	80.7%	91.3%
IV	74.3%	69.2%	86.2%	81.7%
RD	91.5%	89.1%	93.6%	97.1%
RCT	85.5%	81.3%	92.7%	96.2%
Lab Experiment	91.6%	90.8%	92.7%	96.2%
Identification Strategy	57.9%	53.1%	62.4%	75.0%
Structural Model	66.5%	71.8%	66.1%	61.5%
Administrative Data	72.4%	67.1%	74.3%	85.6%

Notes: Agreement rates between keyword-based and LLM-based classification, stratified by field. Field classification uses NBER program affiliations for working papers and journal identity for published articles.

As a cross-model check, Table 7 compares the two LLMs’ classifications directly: positive rates are within one to two percentage points for seven of the nine categories, with cross-model agreement of 85–99.6 percent (κ between 0.57 and 0.94); the models diverge mainly for event studies (7.2 vs. 13.2 percent of papers flagged) and administrative data (39.6 vs. 48.9 percent).

Compared to the GPT-4o-mini benchmarks reported in Garg and Fetzer (2025), the keyword approach achieves comparable accuracy at near-zero computational cost (Figure 14). For tracking broad trends over time—the objective of this paper—simple keyword matching is a reliable tool.

Table 7: Cross-model comparison of LLM classifications

Method	Qwen share	Haiku share	Agreement	κ
DiD	12.1%	11.5%	98.0%	0.90
Event Study	7.2%	13.2%	93.2%	0.63
IV	7.5%	9.3%	97.6%	0.84
RD	4.9%	5.3%	99.3%	0.93
RCT	9.3%	9.1%	98.7%	0.92
Lab Experiment	3.3%	3.7%	99.6%	0.94
Identification Strategy	21.9%	22.4%	85.1%	0.57
Structural Model	31.7%	32.3%	92.5%	0.83
Administrative Data	39.6%	48.9%	85.3%	0.71

Notes: Share of papers each model classifies as using each method, among the 750 sampled papers classified by both models (Qwen3.6-35B-A3B-FP8 and claude-haiku-4-5-20251001, temperature 0). Agreement is the share of papers where the two models’ binary flags coincide; κ is Cohen’s kappa.

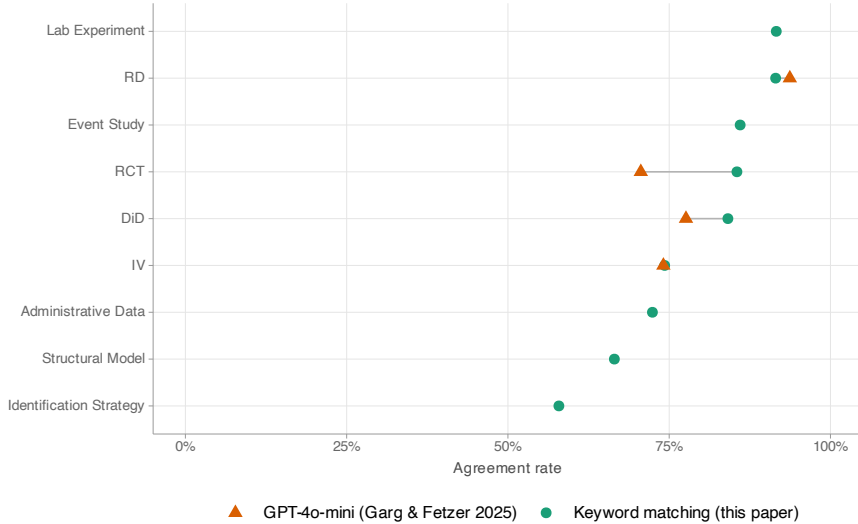


Figure 14: Agreement rates between keyword matching and LLM classification, by method. Where available, GPT-4o-mini accuracy from Garg and Fetzer (2025) is shown for comparison.

Benchmark against hand-coded classifications

As a complementary validation exercise, I benchmark the keyword approach against the hand-coded classifications in Brodeur, Cook, and Heyes (2024), who had research assistants read and classify 1,108 published articles into four categories: DID, IV, RCT, and RDD. Matching papers by title yields 484 overlapping articles.

Table 8 reports the results. Keyword matching achieves high recall for IV (97%) and RDD (95%), meaning it detects nearly all papers that trained coders identify as using these methods. Recall for DID is 97% and RCT recall is 89%. Precision is in the 62–69% range for DID (69%), RDD (65%), and RCT (62%); IV is lower at 57%, reflecting how broadly IV terminology is used in robustness checks and secondary specifications.

An important caveat when interpreting these precision numbers: the Brodeur et al. benchmark assigns each paper a single *primary* method, while keyword matching flags *any* method discussed in the paper. These are fundamentally different classification tasks. Among the 116 papers that keywords flag for IV but Brodeur et al. classify under a different primary method, 53 are coded as DID, 42 as RCT, and 21 as RDD. Many of these papers likely do discuss or employ IV—as a robustness check, a complementary strategy for handling noncompliance in an RCT, or a secondary specification—even though it is not the primary research design. Similarly, of the 71 DID “false positives,” 32 are papers whose primary method is IV but that also employ difference-in-differences in some capacity. What looks like low precision against a single-label benchmark may partly reflect the multi-method reality of modern empirical economics.

The pattern of errors differs across methods in a predictable way. For IV, keywords achieve high recall (97%) but moderate precision (57%): nearly every IV paper is detected, but many papers that use IV as a secondary strategy are also flagged. This will overstate the *level* of IV adoption. For RCT, the pattern is somewhat different—lower precision (62%) and high recall (89%)—because the broad keyword patterns for experiments capture many papers that discuss experimental methods without using them as a primary design. Notably, of the 11 RCTs that keywords miss, 8 are flagged for identification strategy and 3 for IV (likely intent-to-treat designs with IV for compliance)—the papers are detected, but under a different methodological label.

These biases in levels need not bias the *time trends* that are the focus of this paper, provided that the error rates are approximately stable over time. A constant false-positive rate for IV would shift the trend line up without changing its slope. The main risk is that terminology conventions themselves evolve—for instance, if “event study” increasingly appears in DiD papers over time, the false-positive rate for event studies would rise, potentially conflating a terminological shift with a methodological one. The DiD decomposition analysis in Appendix C addresses this concern directly by separating strict DiD from event-study language.

For the purpose of tracking *diffusion* of methods across fields over time, the mention-detection approach has a natural appeal. A paper that employs DID as its primary identification strategy and IV as a robustness check reflects adoption of *both* methods. The keyword approach captures this; a single-label classifier does not. At the same time, keyword matching cannot distinguish between

Table 8: Keyword Classification vs. Brodeur et al. (2024) Hand-Coded Benchmark

Method	N_{hand}	N_{kw}	Precision	Recall	F1	Accuracy	κ
DID	160	226	68.6%	96.9%	80.3%	84.3%	0.679
IV	157	268	56.7%	96.8%	71.5%	75.0%	0.518
RCT	104	151	61.6%	89.4%	72.9%	85.7%	0.637
RDD	63	93	64.5%	95.2%	76.9%	92.6%	0.727
Overall	484	738	62.3%	95.0%	75.3%	—	—

Notes: Benchmark against hand-coded classifications from Brodeur, Cook, and Heyes (2024). N_{hand} = papers classified as using this method by Brodeur et al.; N_{kw} = papers flagged by keyword matching. Precision = share of keyword-flagged papers confirmed by hand coding; Recall = share of hand-coded papers detected by keywords. Brodeur et al. assign a single method label per article, so a keyword flag for a genuinely-used secondary method is counted as a false positive; precision is therefore conservatively deflated. Sample: 484 papers matched by title across both datasets.

a paper that *uses* a method and one that merely *cites* it in a literature review, which is the main source of false positives. The time-series implications depend on whether the ratio of use-mentions to cite-mentions is stable over time—a plausible assumption for established methods, though less certain for methods undergoing rapid diffusion.

Comparison with Garg and Fetzer (2025)

A final comparison uses the released LLM classifications from Garg and Fetzer (2025), available [here](#). This comparison uses their *method classifications*, not their headline share of causal claims. Garg et al. use LLMs to extract causal claims as graph edges and assign each edge a method label. The method labels are mutually exclusive at the edge level: a given extracted edge is classified as DID, IV, RCT, RDD, event study, theoretical/non-empirical, and so on. They are not mutually exclusive at the paper level, because different edges in the same paper can receive different method labels. This makes their data naturally comparable to the multi-label keyword measures used here.

For an apples-to-apples method comparison, I aggregate the edge-level method labels to the paper level. A paper is flagged as using method m if at least one extracted edge is assigned method m .¹¹ Matching on NBER working paper numbers yields 26,100 papers observed in both datasets. One note on terminology: the “Exp./Quasi-exp.” rows in Tables 9, 11 and 12 use the narrow Garg-compatible definition (DiD, event studies, RCT, RDD) and *exclude* IV, while the main-text Exp./QE measure includes IV (alongside lab experiments and bunching). Table 9 also reports an “Exp./QE + IV” row that maps more directly to the main-text measure.

Table 9 reports paper-level agreement between the keyword flags and the Garg et al. method flags. Agreement is high for every method, ranging from 76% for the broad experimental/quasi-experimental-plus-IV composite to 96% for regression discontinuity. The composite experimental/quasi-experimental measure has an agreement rate of 80% and a binary correlation of 0.48; adding IV

¹¹I also require that the method label appears in at least four of their nine independent LLM runs, as they do in their paper.

raises the correlation to 0.51. RCT agreement is especially strong: although keywords flag more experimental papers than Garg et al. (10.3% versus 4.7%), the two indicators have a correlation of 0.55. The largest level difference is IV, where keywords flag 23% of papers and Garg et al. flag 7%, consistent with IV terminology appearing frequently in robustness checks, method discussions, and literature reviews.

Table 9: Paper-Level Agreement: Keyword Flags vs. Garg and Fetzer (2025) LLM Method Classification

Method	Agreement	r	Keyword Pos. Rate	Garg Pos. Rate	Both Flag
Strict DiD	86.0%	0.447	13.1%	16.3%	2,006
Event Study	93.9%	0.377	7.2%	1.7%	357
DiD (composite)	84.6%	0.457	16.5%	17.7%	2,445
IV	84.0%	0.496	22.8%	7.1%	1,821
RDD	96.5%	0.490	4.5%	2.2%	411
RCT	93.1%	0.549	10.3%	4.7%	1,040
Exp./Quasi-exp.	79.7%	0.478	28.3%	23.8%	4,152
Exp./QE + IV	76.3%	0.509	41.9%	29.9%	6,270
Structural Model	84.3%	0.246	16.8%	3.4%	581

Notes: Computed over $N = 26,100$ matched NBER working papers. For paper i and method m , let $K_{im} = 1$ if the keyword search flags paper i for method m , and $G_{im} = 1$ if at least one Garg and Fetzer (2025) LLM-extracted edge in paper i is classified as method m with edge overlap ≥ 4 (the method label appears in at least 4 of 9 independent LLM runs). Agreement is $N^{-1} \sum_i \mathbf{1}\{K_{im} = G_{im}\}$, the share of papers on which the two binary classifications match, including both joint positives and joint negatives. r is the Pearson correlation between K_{im} and G_{im} (the phi coefficient for two binary variables). Keyword Pos. Rate is $N^{-1} \sum_i K_{im}$. Garg Pos. Rate is $N^{-1} \sum_i G_{im}$. Both Flag is $\sum_i K_{im} G_{im}$, the number of papers flagged by both measures. “DiD (composite)” includes strict DiD or event studies for both measures. “Exp./Quasi-exp.” includes DiD, event studies, RCT, and RDD.

Table 10 compares both approaches against the hand-coded benchmark in Brodeur, Cook, and Heyes (2024), but the comparison is not on an identical matched sample. The keyword benchmark matches 484 articles to the corpus used here (385 to published journal versions and 99 to NBER working-paper versions). Garg et al. match 285 papers because their corpus consists of NBER working papers and CEPR discussion papers, so they must find working-paper versions of the published articles coded by Brodeur et al. The smaller Garg et al. sample therefore reflects corpus coverage, not a weaker title-matching procedure.

Within each matched sample, the results show the same precision–recall trade-off seen above. Keywords are recall-oriented: they recover 97% of DiD papers, 97% of IV papers, and 95% of RDD papers in the Brodeur benchmark, but precision is lower because the keyword flags include secondary uses and mentions. The Garg et al. method classifier is more conservative: precision is higher for DiD, IV, and RDD—especially IV, where precision is 93%—but recall is lower for each of these methods. F1 scores are consequently similar, with the keyword measure slightly higher in this benchmark because its recall advantage is large. RCT is omitted from the Garg et al. column because only two of their Brodeur-matched papers are labeled RCT, making precision and recall uninformative; this is a base-rate issue in the matched benchmark sample, not evidence that their

RCT classifier performs poorly in the full corpus.

Table 10: Brodeur et al. Benchmark: Keywords vs. Garg and Fetzer (2025) LLM Classification

Method	Keywords			Garg et al. LLM		
	Prec.	Recall	F1	Prec.	Recall	F1
DiD	68.6%	96.9%	80.3%	73.8%	67.8%	70.7%
IV	56.7%	96.8%	71.5%	92.6%	53.4%	67.7%
RCT	61.6%	89.4%	72.9%	<i>n/a (base rate < 1%)</i>		
RDD	64.5%	95.2%	76.9%	73.2%	75.0%	74.1%

Notes: Both approaches benchmarked against hand-coded classifications from Brodeur, Cook, and Heyes (2024). Keyword results use 484 matched papers; Garg et al. results use 285 matched papers at their EO ≥ 4 threshold. RCT metrics are omitted for Garg et al. because only 2 of 285 matched papers are labeled RCT by Brodeur et al., making precision/recall estimates uninformative.

Finally, Tables 11 and 12 show that the two approaches give the same substantive account of the credibility revolution. In both measures, Applied Micro has the highest experimental/quasi-experimental share, followed by Finance and then Macro/Others. The levels are also close: for the composite experimental/quasi-experimental measure, the field-specific keyword and Garg et al. rates differ by less than two percentage points in finance and macro/other, and by about seven percentage points in applied micro. The time-series patterns are similarly aligned. Both measures show a monotonic rise in experimental/quasi-experimental methods from 2000–2004 to 2020–2024, and both identify DID and RCTs as the categories driving much of the post-2000 increase. The remaining discrepancies are informative rather than destabilizing: Garg et al. classify more papers as DID, suggesting that LLMs recognize designs described without standard keyword phrases, while keywords classify more papers as IV, reflecting the broad use of IV terminology outside primary research designs.

Table 11: Field-Level Method Shares: Keywords vs. Garg and Fetzer (2025) LLM Classification

Field	Exp./Quasi-exp.		DiD (composite)		IV		RCT		RDD		<i>N</i>
	KW	Garg	KW	Garg	KW	Garg	KW	Garg	KW	Garg	
Applied Micro	35.0%	28.0%	19.5%	19.5%	26.2%	8.5%	14.2%	6.6%	6.2%	2.8%	17,364
Finance	24.5%	22.8%	18.9%	20.3%	16.7%	5.6%	3.8%	1.3%	2.5%	1.6%	4,213
Macro/Others	19.7%	18.1%	13.0%	15.5%	19.3%	6.0%	5.1%	1.9%	2.3%	1.1%	12,495

Notes: “KW” = keyword positive rate; “Garg” = Garg et al. LLM positive rate (EO ≥ 4). Field membership is non-exclusive; a paper with programs in multiple fields contributes to each field.

Table 12: Time Trends: Keywords vs. Garg and Fetzer (2025) LLM Classification

Period	Exp./Quasi-exp.		DiD (composite)		RCT		<i>N</i>
	KW	Garg	KW	Garg	KW	Garg	
2000–2004	13.3%	12.8%	8.9%	11.1%	2.8%	1.0%	3,016
2005–2009	19.4%	21.0%	11.0%	17.4%	5.9%	2.2%	4,319
2010–2014	30.6%	25.5%	16.4%	18.7%	11.8%	5.4%	4,851
2015–2019	39.8%	32.3%	23.2%	22.4%	15.5%	7.4%	5,407
2020–2024	45.6%	32.6%	28.0%	22.3%	17.4%	8.3%	5,038

Notes: Five-year averages for matched NBER papers published 2000 or later. “KW” = keyword positive rate; “Garg” = Garg et al. LLM positive rate ($EO \geq 4$).

B Graphical Revolution

Figure 15 plots the graphical revolution—the ratio of figure mentions to table mentions—overall and by field. The trend continues upward across all fields, consistent with Currie, Kleven, and Zwiers (2020b). Macro/other generally has the highest ratio, though applied micro has tracked above finance in recent years and the ordering is not stable across the full sample period.

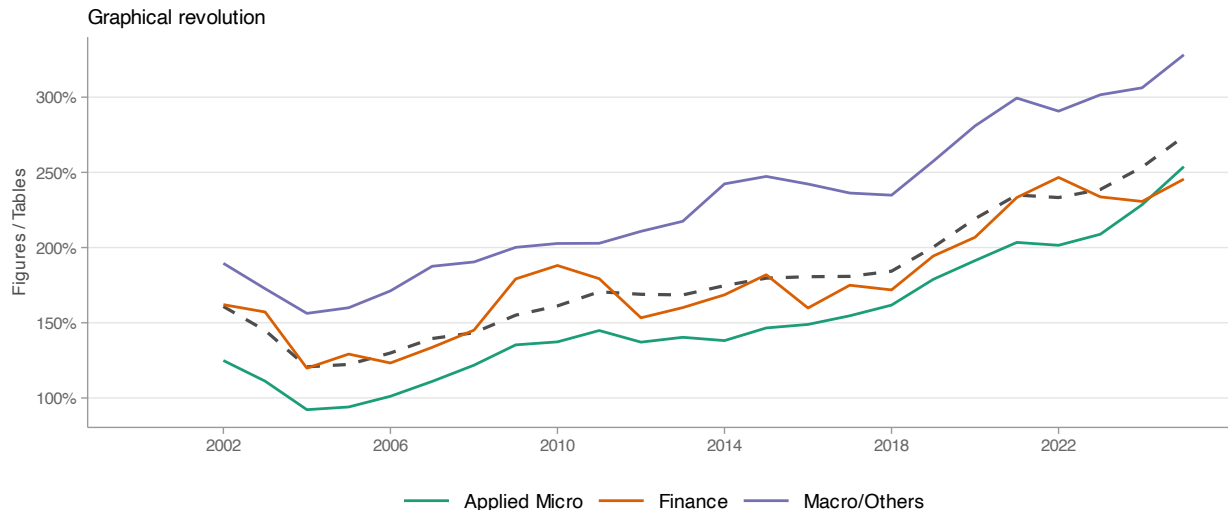


Figure 15: Graphical revolution trends in NBER working papers (three-year moving averages). Colored lines show field-specific trends; dashed black line shows the overall aggregate.

C DiD Decomposition: Strict DiD vs. Event Studies

The main text uses a composite “DiD” measure that flags papers mentioning either “difference-in-differences” (strict DiD language) or “event study.” This appendix decomposes the two components.

Figure 16 shows the decomposition by field. Panel (a) plots the strict DiD measure (excluding event studies), while panel (b) plots any event study mention. Panels (c) and (d) focus on two

subsets: papers mentioning event studies *without* strict DiD language (panel c), and—within that subset—papers that also mention “abnormal return,” a strong indicator of a financial event study rather than a DiD-style event study (panel d). In finance, a substantial share of the “event study only” papers mention abnormal returns, consistent with the concern that the composite DiD measure captures some financial event studies that are methodologically distinct from difference-in-differences designs.

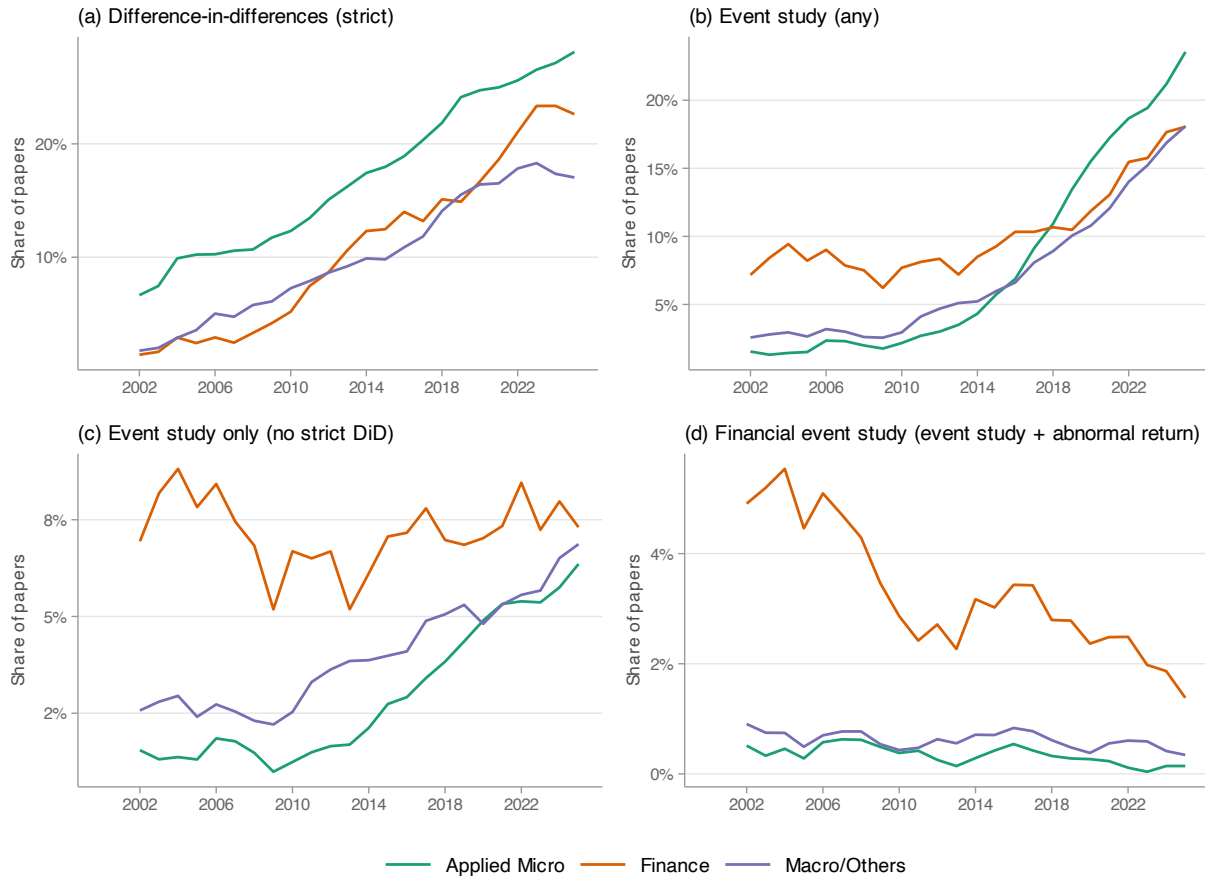


Figure 16: Decomposition of the DiD measure by field. Three-year moving averages. Panel (a): strict DiD language only. Panel (b): any event study mention. Panel (c): event study mentions without strict DiD language. Panel (d): event study mentions with “abnormal return” (financial event study proxy).

D Exclusive Field Classification

The main analysis uses non-exclusive field labels: a paper that lists both applied micro and finance programs counts in both categories. This appendix re-runs the main results using an exclusive classification where each paper is assigned to a single field based on its program affiliations. Papers that list programs from multiple fields are excluded.

Under exclusive classification, the sample comprises 18,697 papers (11,828 Applied Micro, 1,758

Finance, 5,111 Macro/Others). Figure 17 replicates the main time-series analysis. The cross-field gaps are qualitatively similar and, if anything, slightly wider under exclusive classification—consistent with cross-listed papers pulling the non-applied-micro averages toward applied micro.

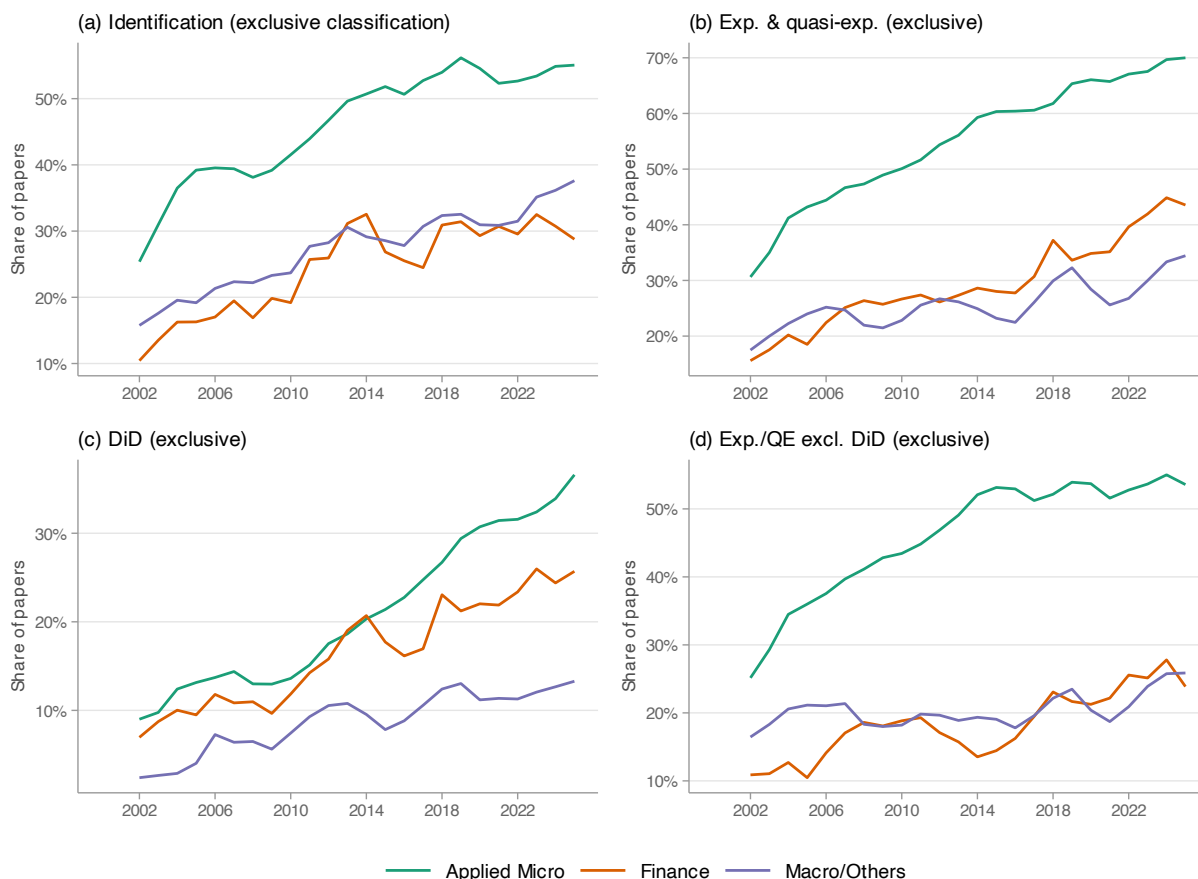


Figure 17: Credibility revolution trends under exclusive field classification. Only papers whose programs all map to a single field are included. Three-year moving averages.

E Rate of Change across Fields

The main text compares *levels* of credibility revolution method mentions across fields. Figure 18 plots the *rate of change*—the first difference of each field’s share, smoothed with a three-year moving average. If all fields are converging to the same steady state, the fields with lower levels should show higher growth rates. If fields have different steady states, growth rates should be similar.

The data are more consistent with different long-run equilibria than with simple convergence. Applied micro’s growth rate in identification (panel a) was near zero from 2016 through the early 2020s, while finance and macro showed positive growth over the same period—though the most recent years show some convergence in growth rates. For experimental/quasi-experimental methods (panel b), all three fields show positive growth, with finance slightly above macro.

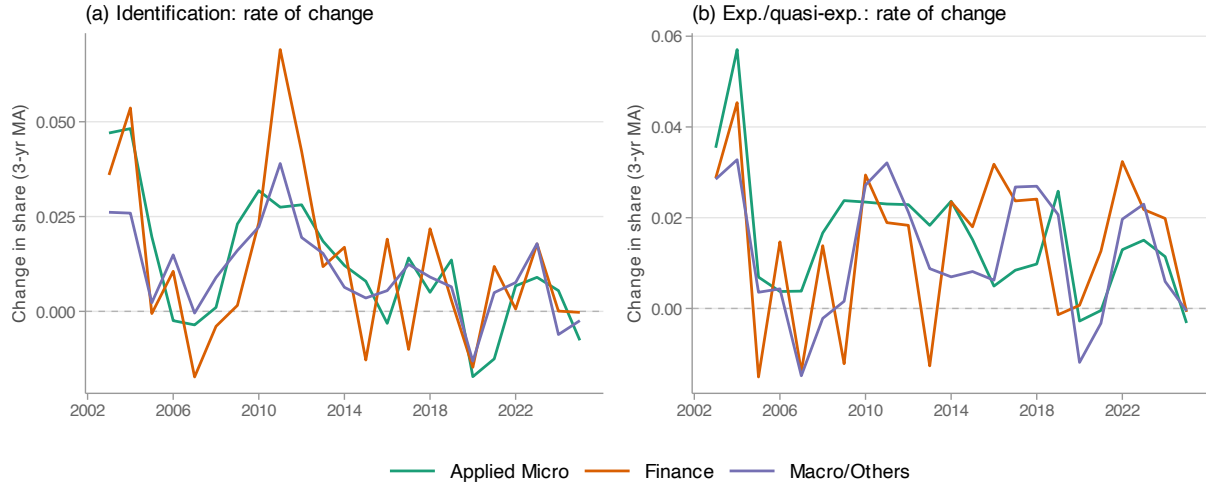


Figure 18: Rate of change in credibility revolution measures by field (three-year moving average of first differences).

F Structural Model Measure: Broad vs. Narrow

The main text uses a broad structural model measure that includes GMM and MLE keywords. Because GMM and MLE are used in many non-structural contexts (e.g., MLE for logit/probit, GMM for moment conditions in reduced-form panel models), this may overstate the share of truly structural papers. Figure 19 compares the broad measure (panels a, c) with a narrow measure that excludes GMM and MLE (panels b, d).

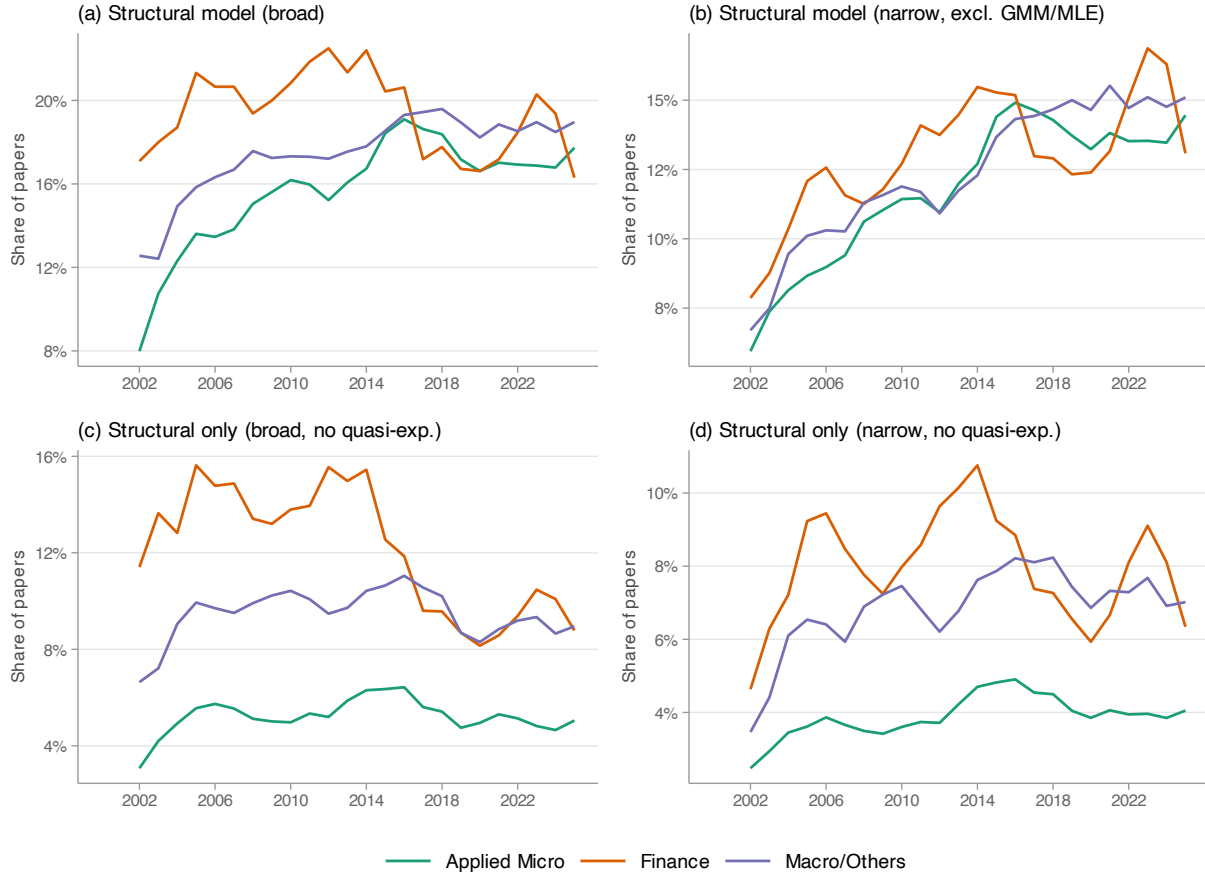


Figure 19: Structural model measures: broad (including GMM/MLE) vs. narrow (excluding GMM/MLE). Panels (a) and (b): all papers. Panels (c) and (d): papers that do not also mention experimental/quasi-experimental methods.

G Denominator Composition

The main analysis computes method shares as a fraction of *all* papers in each field. However, finance and macro/other have a larger share of purely theoretical papers that would never use quasi-experimental methods, potentially inflating the measured cross-field gap. To assess whether denominator composition drives the results, I restrict the sample to “empirical” papers—defined as those that mention at least one of: identification, any experimental or quasi-experimental method, administrative data, survey data, or structural estimation. This is a minimal bar: a paper that discusses none of these is likely theoretical.

In 2024, 83 percent of applied micro papers meet this threshold, compared to 73 percent in finance and 72 percent in macro/other. Figure 20 replicates the method-specific time-series analysis (Figure 4) for both all papers (left column) and the empirical subsample (right column). The cross-field gap narrows—from 65/51/47 percent (unconditional) to 78/69/66 percent (conditional) for experimental and quasi-experimental methods in 2024—but persists across all method categories.

Denominator composition accounts for part of the cross-field gap; the remainder reflects differential adoption of credibility revolution methods within empirical work.

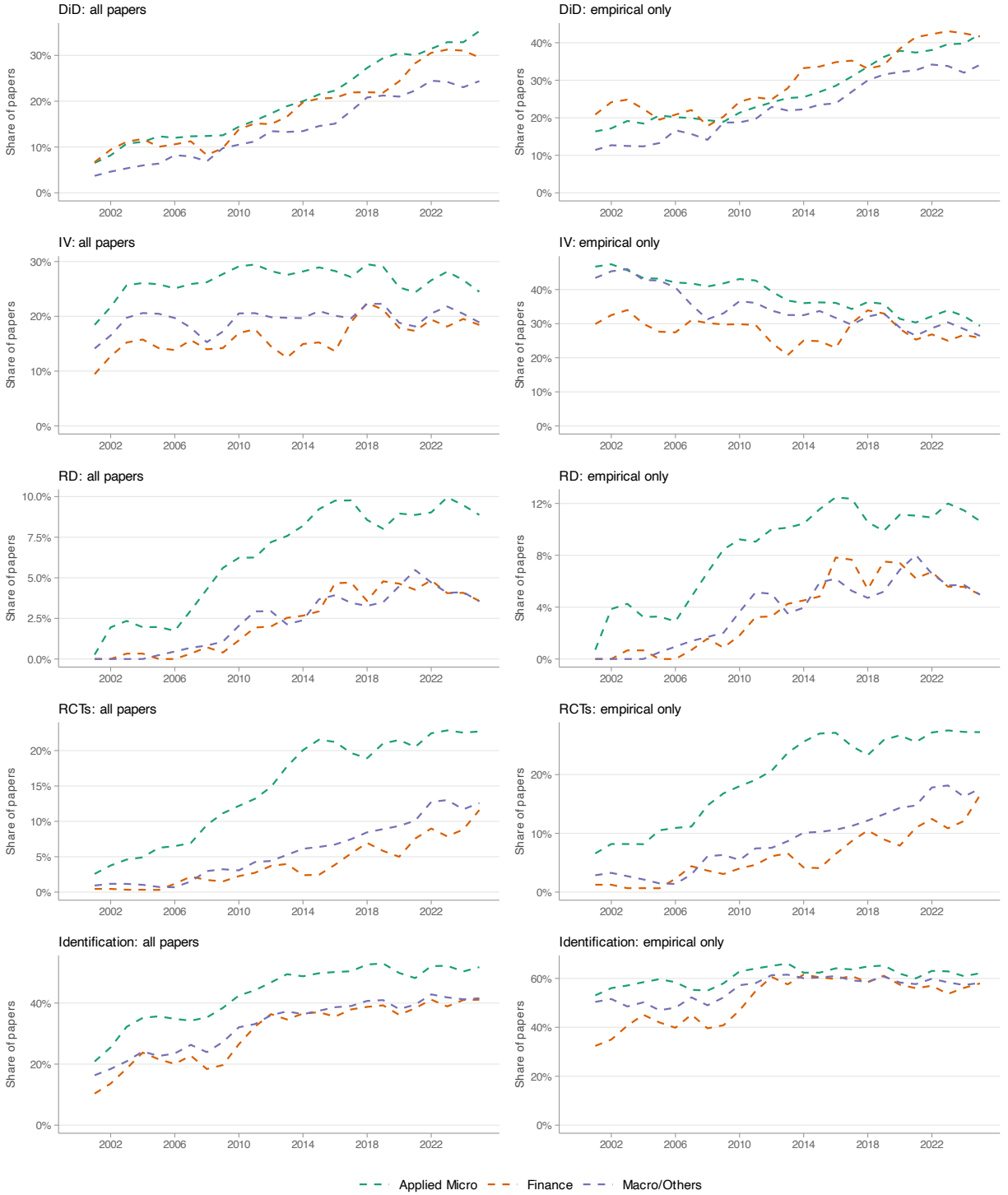


Figure 20: Method-specific trends by field: all papers (left) vs. empirical papers only (right). Empirical papers are defined as those mentioning at least one of: identification, experimental/quasi-experimental methods, administrative data, survey data, or structural estimation. Two-year moving averages.

H Synthetic Control and Synthetic DiD

Figure 21 plots synthetic control and synthetic DiD mentions by field in separate facets. The left facet confirms the decline in synthetic control mentions after 2020, which is evident in applied micro and macro/other, though finance shows a slight increase from a near-zero base. The right facet shows that “synthetic DiD” (including “SDID” and “synthetic difference-in-differences”) has emerged but remains rare, suggesting that the decline in synthetic control is not fully explained by substitution toward synthetic DiD methods. Note the different y-axis scales.

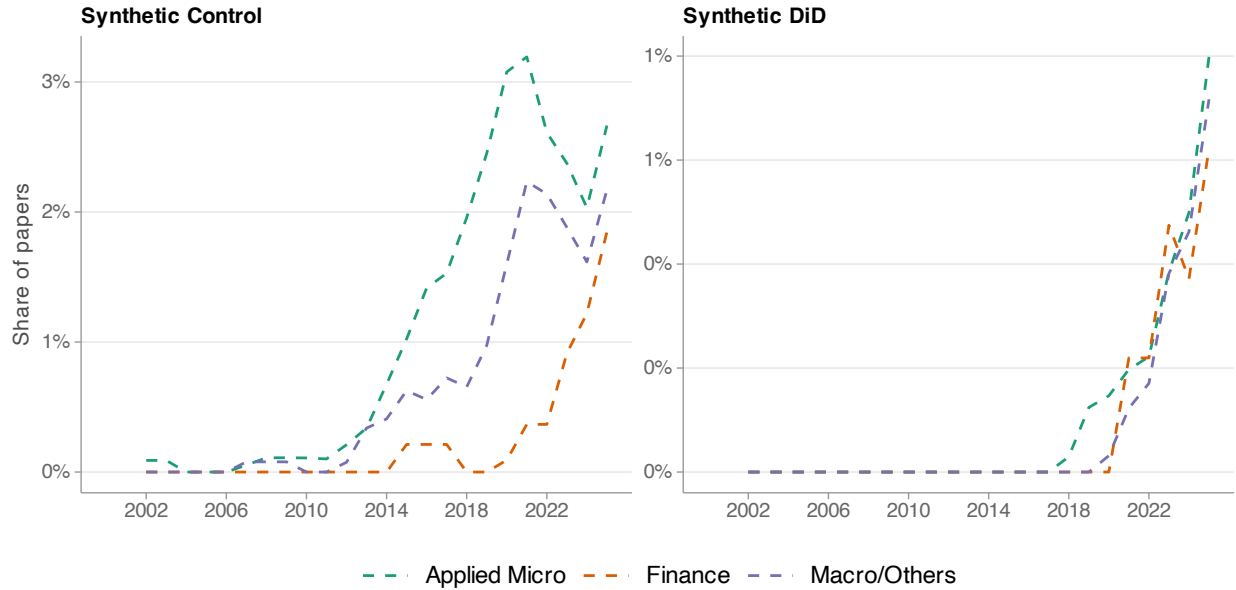


Figure 21: Synthetic control and synthetic DiD mentions by field (three-year moving averages). Note: y-axis scales differ across facets to accommodate different prevalence levels.

I Top Journals: IV and Structural Model Trends

Figure 22 presents the full time series of method mentions across fields in the journal sample (three-year moving averages). The main text shows slope charts for identification and DiD; this figure adds instrumental variables and structural models and shows the annual dynamics. Structural model mentions are markedly higher in finance and macro journals, reinforcing the picture from the working papers that these fields maintain a larger structural modeling tradition alongside the adoption of quasi-experimental methods. IV trends are mostly flat in applied micro and finance, with two exceptions: the macro series rises sharply after 2019, and the *Journal of Econometrics* series trends upward over the sample. Both rises are at low base rates with relatively small samples (especially for macro), and could reflect either substantive movement or sample-composition shifts; the working-paper IV series in the main text is much flatter, so I do not lean on these patterns in the text.

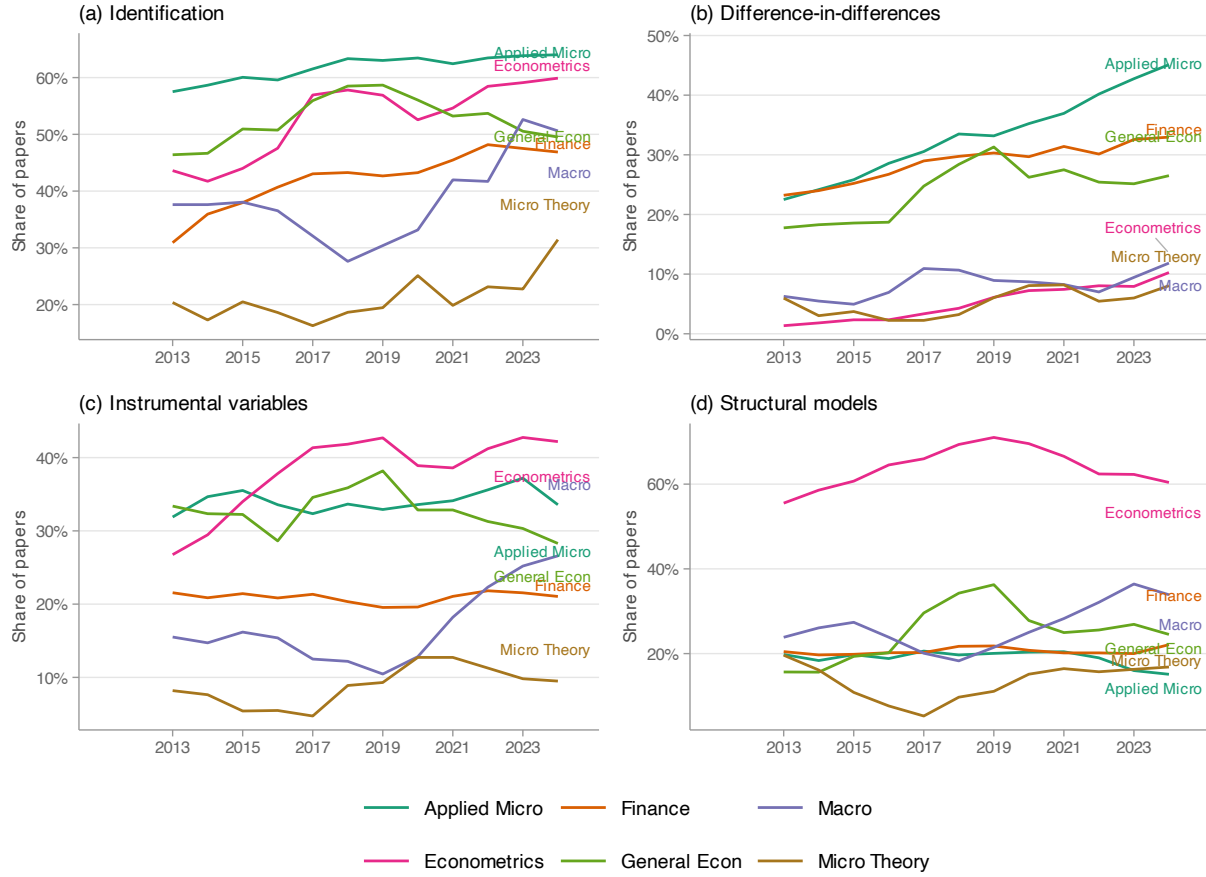


Figure 22: Full time series of method mentions across fields in top journals (2011–2024). Three-year moving averages. Panels (a)–(d): identification, DiD, instrumental variables, structural models. See text for journal-to-field mapping.

J Journal Analysis: Excluding General Economics Journals

The main journal analysis includes general-interest journals (AER, QJE, JPE) classified by JEL codes. Because JEL classification is noisy, this appendix re-runs the journal analysis using only field-specific journals: AEJ Applied and AEJ Policy for applied micro, JF/JFE/RFS for finance, and AEJ Macro for macro. Figure 23 shows that the cross-field patterns hold using only field-specific journals.

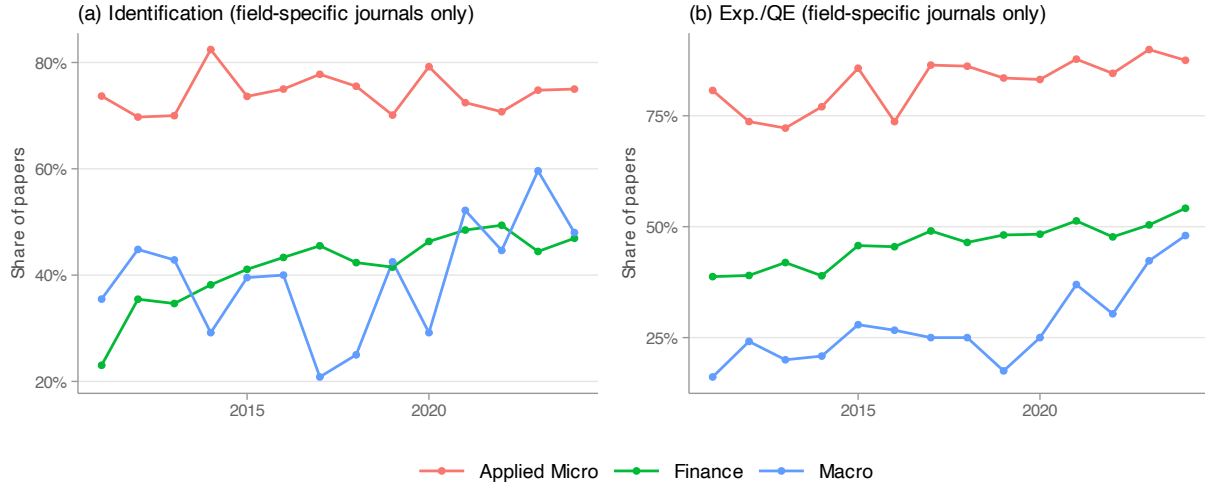


Figure 23: Journal trends excluding general economics journals (field-specific journals only).

K Journal Text Extraction Coverage

Table 13 documents text extraction rates by journal and year. Extracting full text from published journal PDFs is challenging: publisher access restrictions, varied PDF formats, and multi-column layouts all affect extraction quality. Coverage is generally high for journals where PDFs were readily accessible (AEJ journals, AER, finance journals) but incomplete for others. Two journals are excluded from the main analysis due to insufficient coverage: the *Review of Economic Studies* and *Econometrica*.

Table 13: Text extraction coverage by journal and year. Each cell shows the number of papers with extracted text out of total papers in the database. Shaded cells indicate coverage below 80%.

Journal	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	
AEJ Applied	30/37	40/40	47/47	30/30	45/45	37/37	31/31	46/46	44/45	50/50	41/41	63/63	55/55	64/64	28/28	651
AEJ Macro	31/32	30/31	36/36	24/24	43/43	30/30	24/24	32/32	40/40	48/48	46/46	56/56	52/52	50/50	30/30	572
AEJ Micro	38/38	31/31	34/34	44/44	49/49	43/43	41/41	40/40	38/38	34/34	56/56	82/82	65/65	52/52	35/35	682
AEJ Policy	29/30	37/37	44/44	45/45	46/46	39/39	51/52	48/48	54/54	51/51	57/57	63/64	64/64	64/64	48/48	740
AER	260/274	260/264	257/259	255/257	261/262	273/274	244/264	112/113	134/134	120/120	115/115	114/115	95/96	111/111	87/87	2698
J. Econometrics	150/150	136/136	135/135	150/150	195/195	129/129	118/118	159/159	165/165	262/262	164/164	143/143	192/192	175/175	161/161	2434
J. Finance	61/63	93/94	102/105	78/82	100/103	85/130	71/84	87/104	85/103	88/98	81/90	77/86	94/100	95/98	74/89	1271
JFE	139/140	123/125	135/136	114/116	86/87	134/137	112/114	162/164	136/142	141/143	269/273	88/89	80/82	114/120	122/147	1955
JPE	24/36	33/43	31/44	29/46	41/55	33/55	72/97	89/121	66/89	75/97	69/93	86/115	80/115	79/108	73/93	880
QJE	46/47	56/56	36/45	38/46	39/47	45/54	51/52	32/32	39/40	50/51	48/48	42/42	54/54	38/45	38/48	652
RFS	0/114	96/101	95/100	97/98	109/121	97/104	143/147	132/138	143/145	144/147	135/137	94/94	93/95	85/89	86/111	1549

L J. Econometrics: Time Trends in Credibility Revolution Methods

The main text compares term prevalence between the *Journal of Econometrics* and applied journals in cross-section. Figure 24 shows time trends within the *Journal of Econometrics*, separately for theory/estimation terms (panel a) and credibility revolution terms (panel b). While theory terms (nonparametric, bootstrap, Bayesian, machine learning, time series, simulation) remain dominant, credibility revolution methods—particularly DiD, treatment effects, and heterogeneous treatment effects—have grown substantially since 2011, suggesting the gap may be narrowing.

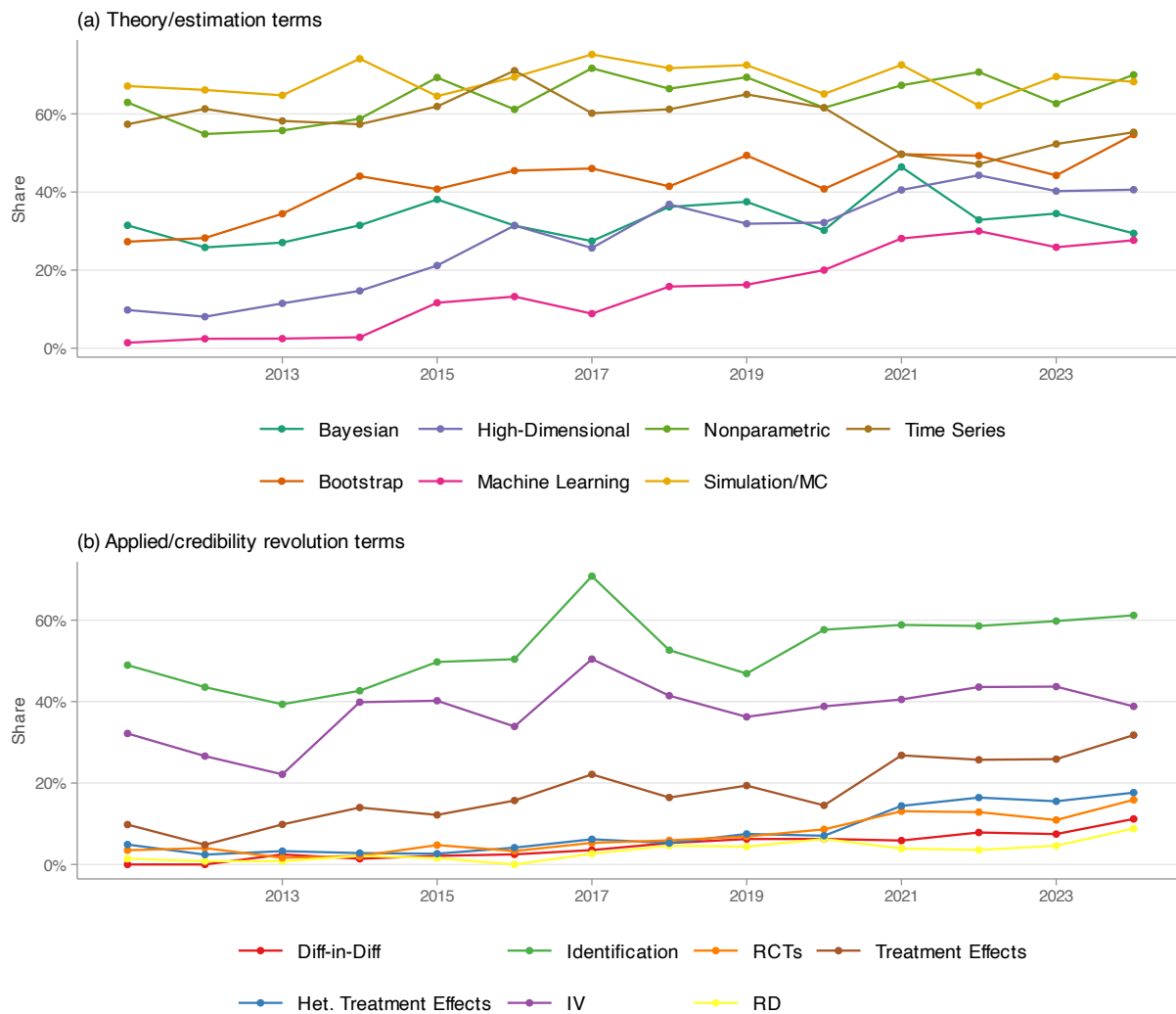


Figure 24: Time trends in the *Journal of Econometrics*, 2011–2024. Panel (a): theory/estimation terms. Panel (b): credibility revolution terms.